

APPROVED METHOD:
INTENSITY DISTRIBUTION MEASUREMENT OF
LUMINAIRES AND LAMPS USING DIGITAL
SCREEN IMAGING PHOTOMETRY
AN AMERICAN NATIONAL STANDARD

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AN AMERICAN NATIONAL STANDARD**

Publication of this report
has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared for IES
By the Testing Procedures Committee**



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CONTENTS

1.0	Introduction and Scope	1
1.1	Introduction	1
1.2	Scope	1
2.0	Background and Basic Principles	1
2.1	Noise in CCD Measurements	2
3.0	Test System Geometry	4
4.0	Angular Resolution	5
5.0	Stray Light	5
6.0	Specification of the Digital Camera	6
6.1	Still vs. Video Cameras	6
6.2	CCD Chip Types	6
6.3	Shutters	7
6.4	Data Access	7
6.5	Dynamic Range	7
6.6	CCD Grade	8
6.7	Camera Controls	8
6.8	Camera Cooling	8
6.9	Lens Selection	8
6.10	Spectral Correction	9
6.11	Neutral Density Filters	9
7.0	Linearity Verification and Useable Range	9
7.1	Dark Field Correction	10
7.2	Response vs. Luminance	10
7.3	Response vs. Exposure Time	11
7.4	Defective Pixels	12

8.0	Calibration.....	12
8.1	Lens Distortion Correction	13
8.2	Geometric Calibration.....	13
8.3	Absolute Intensity Calibration.....	13
8.4	Calibration Correction for Exposure Time	14
8.5	Relative Intensity Calibration.....	15
9.0	Test Procedure.....	15
10.0	Data Reduction and Test Report.....	15
	Glossary.....	16
	Annex A – Geometrical Relationships.....	18
	Annex B – Lens Distortion Correction.....	29
	Annex C – Noise Estimation	30
	Annex D – Linearity Calculation Details.....	31
	References	33

1.0 Introduction and Scope

1.1 Introduction

The intent of this IES document is to describe measurement procedures for the determination of the luminous intensity distribution of certain lamps and luminaires, using a digital camera to capture the projected light distribution from a luminaire onto a screen. Many factors need to be taken into account to achieve accurate digital photometry; this report addresses the most significant of these factors. Because this is a relatively new technique, considerable detail is provided to assist the reader in understanding the principles, requirements, and limitations involved. As more experience is gained and as digital cameras increase in their capabilities, further requirements are likely to be developed. At this stage of development, digital photometry offers very high-speed data collection, as compared to a traditional goniophotometer, although typically over a restricted angular range.

1.2 Scope

This IES Approved Method addresses the use of digital cameras incorporating a CCD array. However, it should be noted that other types of digital camera sensors, such as CID (charge-injection device) arrays, CMOS (complementary metal oxide semiconductor) arrays, and scanned photodiode arrays, could be acceptable for photometry. Requirements for accuracy and the special conditions for this form of light measurement are covered. The numerous factors to be taken into account for hardware selection and software development are described. Calibration requirements are specified, as are data reduction techniques.

Other uses of cameras to perform digital photometry, such as capturing the distribution of a luminaire by aiming the camera at the luminaire, close field photometry techniques, and methods using integrating spheres, are outside the scope of this report.

Requirements that are identical to those for conventional goniophotometry of luminaires and not described specifically in this document include:

- Lamp Selection and Seasoning
- Selection and Preparation of the Test Luminaire

- Luminaire Positioning
- Electrical Supply
- Electrical Instrumentation
- Thermal Environment (other than that for the camera)
- Luminaire Stabilization
- Special Considerations for Particular Lamp Types

These and other items are described in other IES approved methods and are incorporated herein by reference.

2.0 Background and Basic Principles

The measurement of luminaire intensity distributions and associated quantities has traditionally been performed using some form of goniophotometer. Such instruments may use a rotating mirror, a moving photodetector, or a single fixed photodetector with a biaxial goniometer. Each of these systems collects an individual reading and then rotates to the next angular setting, where the process is repeated. Systems using multiple photodetectors have been developed; these increase test speed, as several readings are taken almost simultaneously.¹

In the 1960s, it was shown that photographic techniques could be used to make photometric measurements of the luminance pattern in a space.² A high quality film camera was used with strictly controlled conditions of exposure and film development. The density at any point on the resultant negative image then could be numerically related to the luminance at the corresponding point in the scene, if calibration details were properly addressed. The technique required extensive work to obtain accurate data, as well as sophisticated equipment for the measurement of film density. Even so, it demonstrated the feasibility of a rapid method for the capture of a large amount of photometric data by photographic means.

With the advent of digital cameras in the 1980s, many of the disadvantages of the film technique were eliminated. Photometric methods for the measurement of scene

luminance distribution³ and lamp intensity distribution⁴ were developed using cameras incorporating CCD (charge-coupled device) arrays that replaced the traditional film. Good accuracy was demonstrated despite the relative simplicity of the digital cameras of that time.

Through the 1990s, such cameras grew in their sophistication, as did associated hardware and software, allowing accuracy approaching that of traditional photometry.^{5,6}

The basic operation of a digital camera is similar to that of a film camera. In place of the film, a CCD is placed, which captures the image.⁷ The CCD chip is a flat plane array consisting essentially of rows and columns of extremely small silicon photodetector elements. These individual elements, or pixels, replace the chemical grains of film, and individually respond to the light level incident upon them. Photons falling on an individual silicon element generate a charge, which is captured in an electronic "well." After a momentary camera exposure for a controlled period, the charge deposited in each individual well is downloaded and digitized. The digitized charge value for an individual pixel will bear a linear relationship to the luminance of the corresponding point on the screen, provided proper controls and calibration details are addressed. Under such conditions, rapid and accurate light measurement is possible.^{5,6}

There are two other methods for measuring intensity distributions with a digital camera. One involves pointing the camera directly at the light source, measuring the near-field luminance at many angles, and then ray tracing to calculate the far-field intensity or illuminance at any distance.⁸ The other method is using an imaging sphere, which is not too different in concept from the process described in this document except that the screen is a hemisphere and the system incorporates special optics to enable the camera to see the entire hemisphere.⁹ A combination of a goniometer and an imaging system can be employed. A large section of the test item's light output can illuminate a reflecting wall, which is then measured by an imaging photometer at high resolution. The lamp is then rotated and another large section of the test item's emission is measured with an imaging photometer. In this way, large goniometric maps at high resolution can be obtained faster than could be done using a discrete detector.

Typical applications of photographic photometry have been developed for the measurement of the luminance distribution in interior spaces. The technique can also be used for luminaire photometry, particularly for light distributions where a limited angular range of intensities is to be captured, as for example with stage and studio lights, reflector lamps, and a variety of automotive lighting devices. It is possible to make measurements over a wider range of angles by mounting the test item on a one-axis, or even two-axis, rotating stage and taking multiple exposures. However, it should be noted that many classes of light sources should not be rotated in the vertical plane, as this affects their light output.

The basic laboratory geometry is shown in **Figure 2-1**. The light to be tested is placed on a fixed stand. Mechanisms for aiming adjustments can be provided, but no rotation of the device during testing is required. A screen, painted with diffuse white (spectrally neutral) paint, receives the light distribution. The test light axis is most conveniently aimed perpendicularly to the screen.

A suitable digital camera is located in close proximity to the test light. Its axis can also be aimed to be perpendicular to the screen, although this is not essential; however, it simplifies the conversion of the pixel measurements into intensity data, as described in **Annex A**. Camera lensing is selected such that the entire screen area is viewed by the camera.

The digital camera transmits the image of the screen light pattern to a computer for digitization and processing. This method has the advantage of allowing extremely rapid testing of highly detailed photometric data, although the angular range of the data may be limited.

As with all photometric testing, intensity values calculated from illuminance values will be truly applicable only for the actual test distance used. Care should be taken to ensure that effects such as beam cross-over for narrow beam equipment are avoided.¹⁰

2.1 Noise in CCD Measurements

Photon noise, dark noise, and readout noise are the three primary sources of noise in a CCD camera. It is important to be able to estimate the noise in a given measurement

Approved Method: Intensity Distribution Measurement of Luminaires
and Lamps Using Digital Screen Imaging Photometry

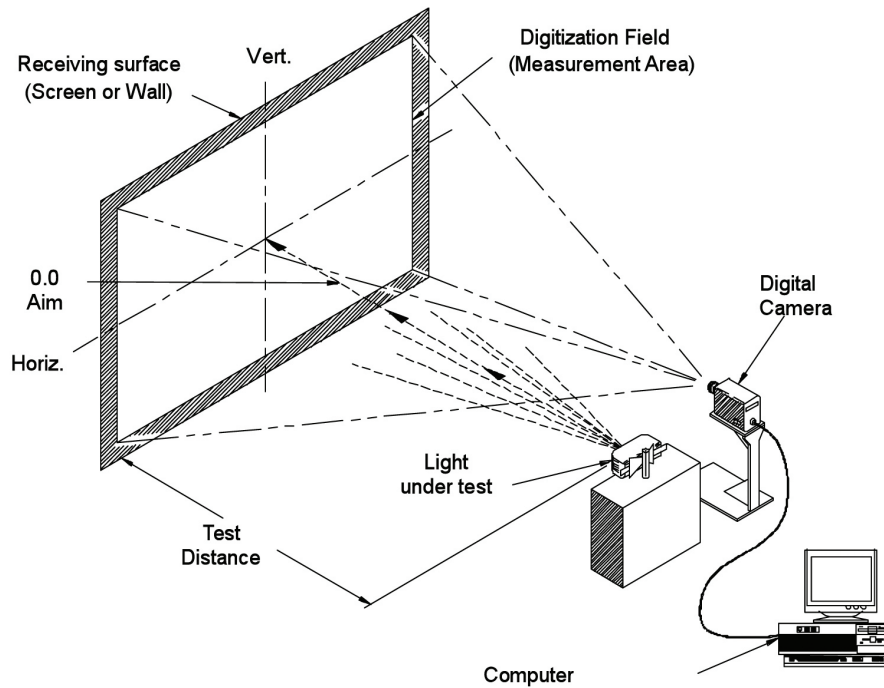


Figure 2-1. Basic laboratory geometry. (© Illuminating Engineering Society)

of pixel charge in order to ensure that appropriate steps are taken to achieve a reasonable signal to noise ratio.

Photon noise arises due to the quantum nature of light. Fluctuations occur in the amount of charge built up in the pixel due to the random arrival time of the photons. This random variation follows a Poisson distribution such that the noise is equal to the square root of the photon-generated charge. This noise is a fundamental property of the type of measurement and is unavoidable. This noise can be reduced by increasing the charge collected on each pixel of the CCD, up to the capacity of the CCD. This capacity is known as the “well depth” or “well capacity” of the CCD and typically varies from 10,000 to 500,000 electrons. CCDs with larger pixels have greater well depth than CCDs with smaller pixels. For CCDs with a well depth of 10,000 electrons, the signal-to-noise ratio would be $10,000/(10,000)^{1/2}$, or 100. A CCD with a 500,000-electron well capacity, the signal to noise would be $500,000/(500,000)^{1/2}$, or 707.

Dark current is thermally generated charge independent of the light incident on a pixel. This process is also quantum in nature. The amount of dark current generated

varies randomly following a Poisson distribution, just as photon-generated charge does. The uncertainty in the dark current, or *dark noise*, is equal to the square root of the dark current. Dark current is often expressed in units of electrons per pixel per second at a given temperature. The average dark current is a constant for a given chip temperature and exposure time. The dark current for a given temperature is proportional to the exposure time. A general rule for the temperature dependence of dark current is that for a given exposure time, dark noise doubles for every increase in temperature of 5 to 6 °C. Therefore, dark noise can be effectively reduced by cooling the CCD.

Readout noise arises from the uncertainties involved in the measurement of the charge in a pixel. The charge in every pixel is converted to an analog signal on the chip itself. This analog signal is then amplified and digitized by electronics off the chip. Readout noise is present in every measurement and increases with readout speed. Readout noise is often expressed in numbers of electrons.

The *total noise* of a pixel measurement is the square root of the sum of the squares of the individual noise components. Since the output of the CCD is given in bits, it is necessary to convert back to the number of electrons. Therefore, the gain of the amplifier that reads the pixel charge is needed. The gain of the amplifier is typically given in units of electrons per ADU, where ADU is analog-to-digital units, or bits. (See **Annex C** for details on estimating noise.)

One method of reducing noise is to make multiple exposures with the same exposure time and average the results. The noise reduction is proportional to the square root of the number of averaged frames. Another method for noise reduction is to average the values of adjacent pixels of an exposure for a single measurement. This technique is permitted, provided the angular resolution requirements of **Section 4.0** are not violated. Adjacent pixel averaging shall occur only after all intensity calibration factors have been applied to the individual pixels. This removes the effect of pixel-to-pixel sensitivity differences.

3.0 Test System Geometry

This document discusses in detail a method that uses a flat screen. It should be noted that a cylindrically shaped screen is also allowed. A curved screen can be utilized especially for cases where a wide field of view is needed and the room size is limited. It should be noted that as the radius of the screen curvature decreases, the stray light reflected from the screen onto itself increases.

This document also discusses in detail a method that requires the camera to be aimed perpendicular to the screen. Another method in use points the camera at the center of the screen. It should be noted that the geometric relationships provided in **Annex A** change if the system uses a non-flat screen with the camera aimed parallel to the axis of the luminaire under test.

The screen surface should be a Lambertian reflector and spectrally neutral. Projection screens should be avoided due to large departures from Lambertian reflectivity.

The maximum angular range of the collected photometric data for a given camera frame is determined by the physical size of the CCD light-sensitive area and the focal length of the camera lens. This in turn may be limited by the size of the screen used and the test distance, which depends on the size of the laboratory. For example, capture of a light distribution over horizontal and vertical angular ranges of $\pm 30^\circ$ using a 5-meter (16.4-ft) test distance will require a screen measuring approximately 6 meters (19.7 ft) square.

For luminaires having a wider intensity distribution than can be conveniently captured on a screen with dimensions suitable for the laboratory space and at the desired test distance, a modified arrangement can be used. For example, the test luminaire may be set on a turntable such that it can be aimed left and right, say by α degrees. Three exposures can be captured for aiming angles of $-\alpha$, 0, and $+\alpha$, and the data can be merged by suitable software. This will extend the horizontal measuring range by up to 2α . It is more convenient, however, to use a screen of sufficient size to capture the entire angular range of interest in a single frame.

In certain cases, measurement of only part of the light distribution from a lamp or luminaire may be the goal of the photometric testing. Consideration should be given to the purpose of the laboratory, and geometry should be selected accordingly.

In designing the laboratory, the desired test distance should be chosen first. The angular range of the desired measurements, horizontal and vertical, is also determined. Screen size is then calculated. The location of the photometric center of the test system shall be defined by the screen-center reference point and the test distance. The following coordinate system allows definition of any point in the test system:

X: The horizontal distance to a given point from the photometric center in a plane *parallel* to the screen. Positive *X* values indicate distances to the right of the photometric center when observed from behind the test luminaire and viewing toward the screen.

Y: The horizontal distance to a given point from the photometric center in a direction *perpendicular* to the

screen. Positive Y values indicate distances from the photometric center toward the screen.

Z : The vertical distance to a given point from the photometric center. Positive Z values indicate an upward direction.

The test luminaire photometric center therefore has coordinates of $X = Y = Z = 0$.

The camera location shall then be selected, ensuring that the camera does not create shadowing of test luminaires on the screen. It shall also be verified that the test luminaire does not block the camera's view of the screen.

The distance between the camera and the luminaire under test should be great enough to allow convenient access to the luminaire. However, the separation distance should not be greater than necessary, as it is critical that the camera's field of view overlap the lighted area to be measured. The following distances shall be measured:

X_{LC} : The distance of separation from the reference photometric center of the luminaire to the center of the camera lens, measured horizontally and *parallel* to the screen.

Y_{LC} : The distance of separation from the reference photometric center of the luminaire to the front of the camera lens, measured horizontally and *perpendicular* to the screen. (For highest accuracy, the distance should be measured to the front principle plane of the lens rather than the lens front. However, its location will usually be unknown. For practical test distances and common lenses, the error introduced is likely to be negligible.)

Z_{LC} : The distance of separation from the reference photometric center of the luminaire to the center of the camera lens, measured vertically.

These distances are used to correct for the offset between the camera and test luminaire location during data reduction. Measurement accuracy shall be at least ± 3 mm, (± 0.1 inch).

Certain applications, notably the measurement of automotive headlamps, are able to use a considerably smaller vertical distance versus horizontal, allowing capture using a rectangular screen.

Geometrical and photometric angle relationships for the test system geometry are detailed in **Annex A**.

4.0 Angular Resolution

The angular resolution of the photometric system shall be developed from the geometric conditions and pixel array information as provided by the camera manufacturer. This shall be performed for both horizontal and vertical resolution by the method described in **Annex A, Section A.7**. This method computes resolution at the center of the field of view where resolution will be poorest. (A given pixel width will subtend a smaller angular range at the edge of the field of view than at the center. Angular resolution therefore improves towards the edge of the field).

The camera horizontal and vertical resolution shall be deemed satisfactory if each does not exceed one-half of the angle between photometric measurement points. For example, if it is desired to measure the photometric distribution in angular steps of 1.0° , angular resolution shall not exceed 0.5° . This ensures sufficiently detailed data for satisfactory interpolation between pixels if a pixel location does not precisely correspond with a desired measurement angle.

It should be noted that this resolution requirement should be considered an absolute minimum. The noise reduction technique of averaging the values of adjacent pixels reduces the effective angular resolution; therefore, this technique cannot be applied when the camera resolution is at the minimum level.

5.0 Stray Light

Precautions shall be taken to prevent stray light affecting the photometric data. This requires that no significant light shall be received on the screen at any measurement point other than directly from the test luminaire. Primary sources of stray light are secondary reflections from the test lamp off the walls, floor, ceiling, or any objects in the room; background illumination from computers; and other light sources. Light baffling and the use of black room

surfaces to minimize stray light are required. Baffling is especially important when emission angles from the lamp or luminaire are much larger than the screen. Baffling should be designed such that no blockage of light from the luminaire to the screen occurs, and so that there is no shielding of the camera's view of the screen.

Secondary reflections from glossy surfaces can greatly affect the measurements. Screening materials shall be flat black, either paint or cloth, having a diffuse reflectance of no greater than 2.0%. Surfaces that exhibit any significant degree of specularity, even if black, may cast images onto the screen.

If the camera is placed closer to the screen than the test luminaire is, (Y_{LC} is positive), baffles shall be placed between the luminaire and camera to ensure no specular reflections occur from the camera onto the screen.

Modeling the test chamber and apparatus in a lighting design and analysis program with realistic screen and baffle material reflectances may prove useful.

If additional room lighting cannot be avoided, its measurement and subtraction from the measurement of a test luminaire is recommended. It should be noted that the measurement of background lighting will have a noise component. Thus, subtracting the background lighting from measurements of a test luminaire will add noise to that measurement. Therefore, it is recommended that the operation of any light sources other than the luminaire under test be avoided.

The contribution of background illumination of the screen can be estimated by taking a measurement of the screen with the test lamp off. Only additional room lighting will be measured by the imaging photometer. This measurement can then be subtracted from the measurement of a test luminaire to eliminate the affect of background illumination.

6.0 Specification of the Digital Camera

It is critical that the type of digital camera to be

used in luminaire photometry be selected to produce accurate data over the required measurement range. The following sections describe items to be considered in camera selection.

6.1 Still vs. Video Cameras

There are two basic classes of digital cameras. The still, or single-frame, camera captures individual images. These may be equipped with multiple forms of exposure control, typically including exposure time and lens aperture (f -stop). The second type, the video camera as used for television, most commonly operates at 30 frames per second. Exposure control may be achieved by the lens aperture and/or automatic shutter control.

Proper exposure control is essential during system calibration and photometric testing. In addition, the length of exposure required will frequently exceed that offered by television-type cameras. For these reasons, the camera used shall be of the single-frame type.

6.2 CCD Chip Types

There are two common types of area array CCD chips. These are full-frame and progressive-scan (interline transfer) chips. The choice of which type to use will depend on the intended photometric application and other factors.

Nearly the entire surface area of a full-frame chip is light sensitive. This gives full-frame chips good sensitivity. This type of chip does require the use of a mechanical shutter to block light from continuing to interact with the sensor during readout.

Progressive scan (interline transfer) chips have masked areas on alternate columns of the sensor for storage. Once an exposure is completed, the charge in the light sensitive pixels is transferred quickly to the pixels in the adjacent masked column. This eliminates the need for a shutter to control exposure. The area on the CCD devoted to readout reduces the area that may be devoted to charge storage; as a result, progressive-scan sensors can typically store fewer electrons per pixel (smaller full-well capacity) than a full-frame CCD sensor can.

6.3 Shutters

There are two types of exposure timing: mechanical and electronic shutters.

Electronic shutters shift charge based on the clock speed of the camera. Exposure timing error is typically less than 1 microsecond. However, readout smear in progressive-scan CCDs will create a practical limit for the minimum exposure for an imaging photometer. Some sensors may be used at exposure times as short as a few milliseconds without noticeable “smear,” where other progressive-scan sensors should be not used for exposure times shorter than 50 milliseconds.

By contrast, *mechanical shutters* can have timing errors when opening and closing. As a result, the exposure times are not reproducible within several to hundreds of milliseconds. If using a mechanical shutter for exposure control, the system should not be used with exposures shorter than 100 times the shutter timing uncertainty. For larger CCD arrays, larger and slower shutters may be required. Larger shutters typically have worse timing characteristics than smaller shutters.

The following is a list of common shutter types:

1. Electronic: These can be uniform, repeatable, and smear-free even at exposures of a few milliseconds. They exhibit zero latency (see **Section 7.3**).
2. Iris: These mechanical shutters expose the center of the CCD longer than the sides and corners and are relatively slow. Therefore, exposure times shall be greater than 1 second, to minimize this effect.
3. Dual blade: These mechanical shutters also expose the CCD center longer, but open and close faster, so exposure times can be 300 to 700 ms, depending on the CCD size. (Larger CCDs require longer minimum exposures.)
4. Rotating slot: This type of shutter can expose all the areas of the CCD uniformly, assuming the speed of the rotation can be maintained accurately. This type of shutter also exhibits zero latency.

Shutter types 1 and 4 are recommended.

6.4 Data Access

The camera shall be capable of determining the charge

associated with each individual pixel, and digitizing the charge into a corresponding bit count, or pixel value. It shall also provide the facility for connection to a data acquisition interface and computer. Examples of such interfaces are Fire Wire (described in ANSI/IEEE 1394¹¹), Ethernet, Camera Link, and USB 2.0. Software shall be available to identify the bit count derived from each individual pixel over the range of interest, in terms of row and column identifying numbers in the pixel array.

6.5 Dynamic Range

The single-exposure dynamic range of the camera and its associated equipment is a function of the capacity of the pixel wells, the read noise, and the resolution of the analog-to-digital converter. Manufacturers of CCD chips generally define the dynamic range of the chip as:

$$(Linear\ full\ well\ capacity)/(read\ noise), \quad (6-1)$$

or:

$$20 \log(Linear\ full\ well\ capacity)/(read\ noise), \quad (6-2)$$

where full well capacity and read noise are measured in number of electrons. The result of **Equation 6-1** is unitless. The units for the result of **Equation 6-2** are decibels.

These formulas are useful in determining what analog-to-digital converter resolution would be suitable for the chip. For purposes of digital photometry, the single-exposure dynamic range shall be defined by:

$$dynamic\ range = (max\ linear\ bit\ count) - (min\ linear\ bit\ count)/(read\ noise/gain) \quad (6-3)$$

Note: Any regions that are not linear shall be excluded. (Refer to **Section 7.2** regarding the maximum and minimum bits counts of the linear range.)

The dynamic range shall be great enough to capture the luminaire intensity range of interest.

There are two methods of using multiple exposures to increase dynamic range. The first method is to simply average many images with the same exposure time. This reduces noise by the square root of the number of images used. The second method is to perform multiple exposures for a given luminaire using different exposure

times, integrating the useable data into a single array of values.^{12,13} Using this technique, the useable dynamic range can exceed 1,000,000:1. However, this method involves the use of an exposure time of sufficient length to capture screen areas of low light level, and will result in over-exposure of areas of high light level. Over-exposure of wells can cause the well charge to spill over to adjacent wells, (causing "blooming" of the image), giving erroneous readings. Due to this problem, the method of using a range of exposure times to extend dynamic range is restricted to sensors with anti-blooming features.

The dynamic range of a detector can also be extended by characterizing the response-versus-luminance function of the detector for nonlinear regions, and using this function to develop linearity correction functions.

For a single exposure, dynamic range is assisted by:

- A large full well capacity
- Low dark noise; this is achieved by short exposure times, detector design, or cooling
- An analog-to-digital converter of sufficient resolution to adequately digitize the well capacity

A nominal maximum dynamic range of 4,096:1 is generally adequate for photometric testing, although this should be verified for given applications. Achieving such a range requires a camera that incorporates an analog-to-digital converter with 12-bit resolution or greater. The converter shall be incorporated within the camera itself rather than in the remote computer.

It is important to note that the useable dynamic range will be less than the maximum dynamic range (see **Section 7**).

6.6 CCD Grade

The sensitivity of pixels to incident light will vary from pixel to pixel. This variation is taken into account in the calibration procedure (see **Section 8.3**). Some pixels, however, may be defective. The number of defects, particularly cluster defects, should be minimized. During initial camera qualification, any defective pixels shall be identified and appropriate correction shall be applied (see **Section 7.4**). A CCD array containing a set of adjacent defective pixels larger than the minimum

angular resolution requirements of **Section 4.0** shall not be deemed suitable for measurements at that level of angular resolution.

6.7 Camera Controls

The camera exposure time shall be under user control, either manually or by software.

Alteration of the lens aperture introduces a variable that can invalidate the system calibration. Repeatability of the imager performance after turning the lens aperture *f*-stop setting shall be validated. Otherwise, a fixed-lens aperture should be selected that will allow satisfactory data capture for luminaires of the type to be measured.

In order to accommodate different levels of luminaire intensities, the exposure time is controlled to ensure that all required data lie in the linear response portion of the camera's range. Further details for setting the camera response are described in **Section 7**.

6.8 Camera Cooling

A CCD pixel, even in the absence of light, will accumulate signal over time due to thermally generated electrons. The rate at which the signal is accumulated increases exponentially with temperature. This thermally generated signal is often referred to as "dark field." This effect may be significant compared to signal levels generated by the light source under measurement. A proper dark field subtraction will greatly improve the measurement accuracy. Dark field is reduced by cooling of the CCD array, most conveniently by electronic Peltier cooling. If at the longest allowed exposures the dark field are not small compared to the reported signal, then cooling of the camera may be required.

6.9 Lens Selection

The digital camera shall be equipped with a lens that allows capture of the entire area of interest of the screen. This will be determined by the lens focal length and the geometric details of the camera. The camera manufacturer may be consulted to determine the lens focal length required for the capture of a certain screen area.

Lenses of shorter focal length (wider field of view) than necessary should be avoided, as this reduces the angular resolution between data points. Such lenses

also are more likely to exhibit vignetting,³ but this effect will be taken into consideration by the recommended calibration procedure. It should also be kept in mind that image distortions typically increase as the focal length of the lens decreases.

6.10 Spectral Correction

The spectral response of the CCD array shall be corrected to that of the CIE $V(\lambda)$ curve, within practical limits. A typical spectral response curve is illustrated in **Figure 6-1**, and can be seen to have significant peaks and valleys. A spectral filter, which may be placed externally in front of the lens or internally between the lens and the CCD array, shall be used to develop an effective spectral response of the entire optical system to closely approximate the $V(\lambda)$ curve.¹⁴ The degree of fit as expressed by f_1' shall be determined. **Figure 6-2** illustrates a combined spectral response using a filter, for a system achieving $f_1' = 3.8\%$. It is recommended that absorptive type filters be used, as reflective thin film filters can cause highly undesirable reflections.

It should be noted that most CCD cameras have an internal infrared filter. It may be possible to remove this filter and replace it with an internal $V(\lambda)$ filter that will also remove infrared radiation.

It should also be noted that $V(\lambda)$ filters have a transmission that varies with the angle at which the light passes through it. If the light path through the filter is not very nearly normal to the filter, different amounts of spectral filtering may be applied over the sensor surface, leading to a variation in f_1' over the camera's field of view.

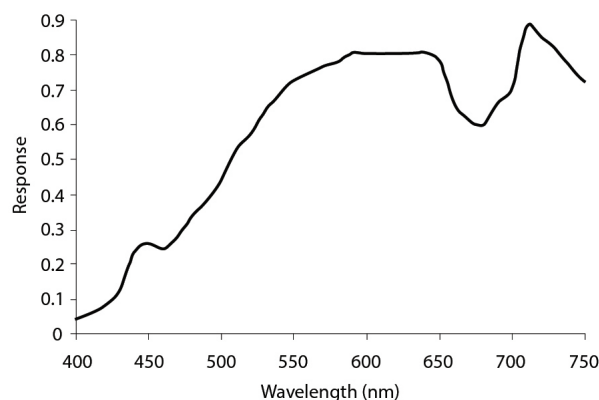


Figure 6-1. Spectral response curve for example CCD. (© Illuminating Engineering Society)

6.11 Neutral Density Filters

Situations can arise where the intensity of the light to be measured is so high that the camera saturates even at the shortest useable exposure time. In this case, the use of a neutral density filter is permitted. The measured intensity values will have to be divided by the transmission factor of the filter in order to correct for the effect of the filter. The uncertainty in the transmission factor of the filter should be small in order to minimize the impact on the uncertainty of the intensity measurement. It is recommended that absorptive type filters be used, as reflective thin film filters can cause highly undesirable reflections.

Since neutral density filters are not perfectly spectrally neutral, they can affect the photopic response of the imaging photometer. It is recommended that caution be used when adjusting the sensitivity range of the imaging photometer because using neutral density filters may change the photopic response of the imaging photometer.

7.0 Linearity Verification and Useable Range

Prior to performing a detailed intensity calibration of the system, the response linearity shall be verified, and the limits of linearity shall be determined. Subsequent use of the system shall be limited to the range of conditions where linearity has been demonstrated to

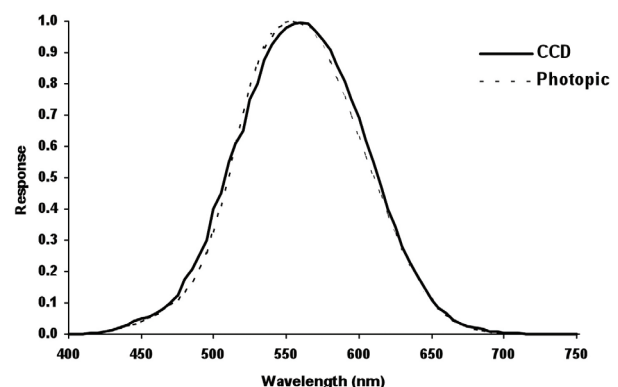


Figure 6-2. Corrected spectral response curve for a CCD and filter combination. (© Illuminating Engineering Society)

lie within $\pm 1\%$. Otherwise, linearity correction shall be applied. Such correction using lookup tables may be supplied by the camera manufacturer.

7.1 Dark Field Correction

Prior to performing linearity evaluation, a technique shall be available to take into account "dark field." *Dark field* denotes pixel readings that are obtained when data are captured in the absence of light. It may be compared to dark current for a conventional silicon photodetector.

Dark field increases linearly with exposure time. The dark field data for a given exposure time are collected in the absence of light striking the CCD. These data can be stored for later use. Dark field measurements need not be made for all possible exposure times. The dark field for a specific exposure time may be interpolated from dark field data with different exposure times. It is recommended that dark field data be collected by averaging a number of identical exposures. This will reduce the noise in the dark field data.

Whenever data are collected for a field of interest, the exposure time shall be noted. The dark-field-corrected reading for each pixel is determined by subtracting the dark field value corresponding to the exposure time from the test reading.

Certain cameras, particularly when used with short exposure times, may produce negligible dark field data, in which case the dark exposure is unnecessary. It should be noted, however, that even high quality cooled cameras may show significant dark field data for long exposures, as may be needed to capture low light levels.

A dark frame with a very short or essentially zero exposure time will not yield zero bits. CCDs are designed this way to prevent negative values. This is referred to as the *bias field*. If no dark field subtraction is performed, then a bias field shall be subtracted. As with dark field, it is recommended that the bias field data be collected by averaging a number of bias frames collected at near-zero exposure times. If dark field data are subtracted, no bias field subtraction is necessary, as the dark field data automatically include the bias field.

7.2 Response vs. Luminance

To quantify response with regard to object luminance, the camera is aimed at a source of variable and known luminance. The luminance is varied from zero to that luminance which produces the maximum attainable bit count. (For example, for a 14-bit system, the maximum bit count would be $2^{14} = 16,384$). At each selected luminance level, the average bit count is computed for a selected group of pixels. The exposure time should be a constant for all luminance levels.

Plotting of the reading (for example, bit count) (after subtraction of dark field readings) versus luminance should yield a curve of the general form illustrated in **Figure 7-1**. CCD pixels exhibit saturation at high bit counts, as evidenced by a flattening of the response curve. The upper limit of the system usable range shall be defined by the bit count at the point where the response curve has deviated 1% from the predicted output at any exposure time allowed. That is, the reported value (e.g., bit count or luminance) will be 1% less than would have been obtained if the response had been perfectly linear.

At low bit counts, values are likely to be affected by system noise, which will be indicated by a lack of repeatability of readings. The individual readings taken at a given luminance level shall be analyzed for repeatability. Each measurement at a given luminance level shall be within $\pm 2\%$ of the average for that luminance level. Bit counts exhibiting a greater variation shall be considered to be outside of the system useable range.

The system usable range shall be defined in terms of bit count, from the point at which $\pm 2\%$ repeatability occurs up to the bit count at which a 1% deviation from linearity occurs.

Data collection or reduction software may be conveniently programmed to reject data outside of the bit count useable range. Where bit counts exceed the useable range, exposure time can be reduced, and the frame can be recaptured. Where bit counts in areas of the intensity pattern that are of interest lie below the useable range, exposure time can be increased to eliminate this problem.

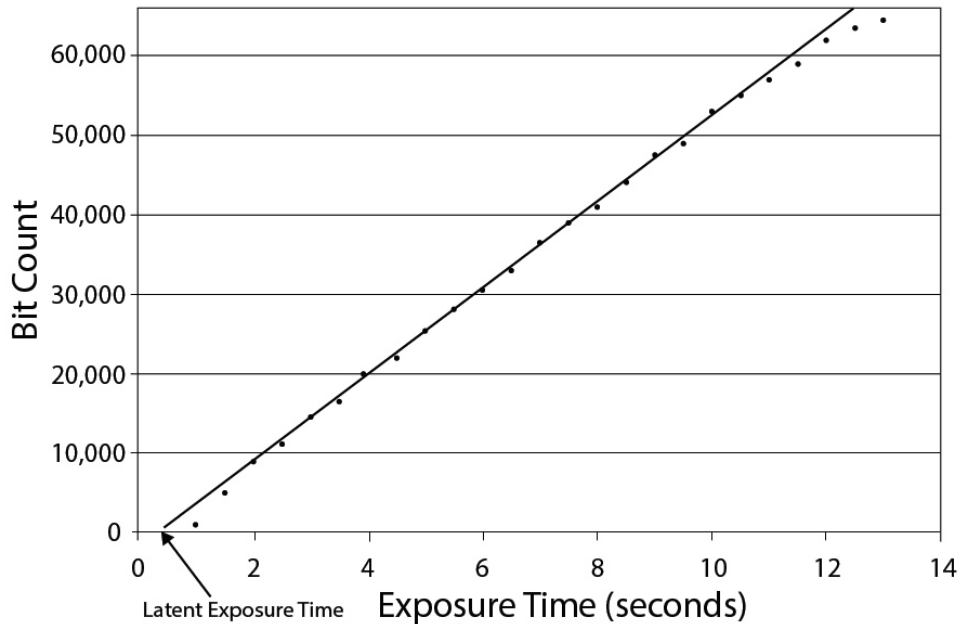


Figure 7-1. Example response curve of bit count vs. exposure time. Note: Departures from linearity are exaggerated here for clarity. (© Illuminating Engineering Society)

It should be noted that factors other than system noise can cause repeated readings to deviate from their average by more than 2%. For example, fluctuations in exposure time can cause bit count variations and typically will affect the entire range of readings. Any such variations, which shall be identified during the described evaluation, shall be reason for rejection of the system.

Another factor that may cause lack of repeatability is flicker of a light source if operated on alternating current (AC). Where possible, direct current (DC) operation is preferred. For light sources using AC, short exposure times shall be avoided; otherwise, light source flicker may affect the data captured. Long exposure times and/or averaging of successive image frames can be used to eliminate any AC effects.

7.3 Response vs. Exposure Time

The response for a given luminance (response vs. luminance) at a constant exposure time is equivalent to the response for a given exposure time (response vs. exposure time) at a constant luminance. It is not necessary to measure response vs. exposure time unless a shutter that exhibits latency is used.

To measure response vs. exposure time, the camera is aimed at a source of variable and known luminance. An exposure time greater than the minimum exposure time dictated by the shutter characteristics is used, and the average bit count of a selected group of pixels is determined. The luminance of the light source may be adjusted to achieve a desired minimum average bit count. The number of pixels averaged should be a sufficient number to achieve repeatability within $\pm 0.5\%$ (see **Annex C**). No defective pixels should be a member of the selected group (see **Section 7.4**).

The exposure time shall be increased to produce a bit count near the maximum attainable, and the bit count is again recorded. Further testing shall be conducted using intermediate exposure times. The bit counts at a given exposure time, after subtraction of dark field counts, shall be averaged. All averages shall be computed with the same set of pixels.

Plotting of the bit count (after subtraction of dark field readings) vs. exposure time should yield a curve of the general form illustrated in **Figure 7-1**. CCD pixels exhibit saturation at high bit counts, as evidenced by a flattening of the response curve. The limits of the system usable range shall be defined on both the high and low ends of the digitizer range by the bit count at

the point where the response curve has deviated 1% from linear; i.e., the bit count will be within $\pm 1\%$ of what would have been obtained if the response had been perfectly linear.

The individual readings taken at a given exposure time shall be analyzed for repeatability. Each measurement at a given exposure time shall be within $\pm 2\%$ of the average for that exposure time. Bit counts exhibiting a greater variation shall be considered outside of the system useable range. (Refer to **Annex D** for a more detailed explanation of the linearity analysis.)

The data throughout the range shall be linear to within $\pm 1\%$, incorporating lookup table corrections for linearity, if used.

The graph that is developed (see **Figure 7-1**) may indicate an intercept on the abscissa. This is referred to as the *latent exposure time* and is related to physical timing considerations in the exposure control. Its value shall be noted, as it will later be applied in calibration and testing.

It should be noted that there is no industry standard among camera manufacturers for the reporting of linearity. A commonly employed formula is:

$$\text{Nonlinearity} = \frac{[(\text{max positive deviation}) + (\text{max negative deviation})]}{(\text{max signal})}, \text{ expressed as a percentage}$$

This method used by manufacturers is much less rigorous than the method described above in this section. Camera manufacturers' claims of linearity should be considered with this in mind.

7.4 Defective Pixels

There is the possibility that a defective pixel may correspond to a desired measurement angle, and therefore the camera shall be evaluated for defective pixels.

A defective pixel will exhibit a nonlinear response to luminance or excessive dark-current production. For example, a "hot pixel" will exhibit a high bit count at all times.

The data collected during the verification of linearity vs. exposure time can be used to identify defective pixels. An exposure producing an average bit count approximately 70% of the maximum linear bit count shall be used to identify defective pixels. Any pixel with a bit count that deviates more than 6% of the average shall be classified defective.

A file identifying any defective pixels shall be incorporated into the data collection software. For any defective pixel, the bit count assigned to it for subsequent measurements shall be calculated by weighting the corner pixels at 50% of the pixel values in the adjacent rows and columns. For example:

1 2 1

2 x 2

1 2 1 ,

where x is the unknown value, and numerals 1 and 2 represent the weights given to the surrounding pixels. In other words, there are eight weighting numbers. The weighting number for the four corner pixels is 1, and the weighting number for the four adjacent pixels is 2.

8.0 Calibration

When operated under certain conditions and over a particular range of light levels, the response of each pixel of the CCD array has good linearity. The charge stored for each pixel after exposure is thus directly proportional to the light it receives. Digitization of the charge thus provides a measure of the luminance level of the corresponding point on the screen, although the relationship is affected by a series of variables.

Screen luminance, in turn, will be dependent upon the intensity of the test luminaire, as well as other factors.

The bit count for a given pixel can be converted directly to test luminaire intensity by the method described in **Section 8.3**. Intermediate conversion to screen luminance and consideration of screen reflectance are unnecessary using the intensity calibration technique described in **Section 8.3**.

8.1 Lens Distortion Correction

A camera lens may not produce a perfect image, due to distortions caused by the lens itself. The effects are classed either as *barrel distortion* or *pincushion distortion*, describing the general shape of an image of a square object. These effects are generally insignificant with lenses having a long focal length, (e.g., telephoto lenses), but can be noticeable with wide-angle lenses such as may be needed in luminaire photometry. Methods are available to determine the effect of lens distortion using a correction equation. When certain parameters of the distortion have been determined, the original image shall be reprocessed using the equation to develop a new image. All subsequent operations then use the reprocessed image to determine the photometric quantities of interest. **Annex B** describes the correction technique. (Note: The correction technique in **Annex B** should be considered the minimum level of correction. Other, more sophisticated distortion correction techniques may be used if desired.¹⁵)

8.2 Geometric Calibration

Geometric factors need to be addressed for the development of a meaningful test system. These relate to the separation distance of the luminaire and camera, the relationship between screen locations and luminaire photometric angles, and the relationship between pixel locations and corresponding screen locations.

A temporary cross-hairs target shall be placed on the screen, with the crossover point corresponding to a chosen screen reference center point at $X = 0$, $Y = D$, $Z = 0$. This may be conveniently formed by tightly stretched thin cables. Alternatively, a laser incorporating optics to produce a cross-hairs projection can be mounted on the test stand and aimed perpendicular to the screen at the screen's optical reference center point. It is critical that care be exercised to prevent accidental direct exposure of the eyes of the operator to the laser beam.

The screen shall be imaged using the test camera with the cross hairs in position. From the resultant image, the pixel corresponding to the screen optical center point shall be identified. A coordinate system then can be developed that relates specific screen location to corresponding pixel. A given screen location

(X , Z coordinates) also can be related to a particular horizontal and vertical photometric angle.

This relationship will be dependent upon the photometric coordinate system chosen, i.e., CIE A, B, or C, or a modification thereof.¹⁶ Thus, the photometric angles may be mathematically related to a specific pixel row and column number, allowing a photometric reading from a particular pixel to be identified with its corresponding photometric angles. The derivation of the relationships involved is provided in **Annex A**.

8.3 Absolute Intensity Calibration

This section describes the use of a standard reference source to determine the sensitivity of each pixel. It is based on the use of identical exposure times for the reference source and test luminaire. **Section 8.4** describes the correction required when the test-luminaire exposure time differs from that used for the standard reference source.

Calibration for absolute intensity shall be achieved using a standard reference luminaire. This may take a variety of forms, but conveniently can be a luminaire, which when placed on the test stand will floodlight the entire area of the screen. This floodlight should preferably have an intensity distribution that is free from irregularities such as hot spots and striations, and should have a light distribution that obeys the inverse-square law at the applicable distance. That is, there should be no beam cross-over.¹⁰ The reference standard luminaire shall incorporate an incandescent source for stability. The source shall be powered by direct current, or exposure times shall be sufficiently long to eliminate deviations caused by flicker (see **Section 7.2**).

The reference luminaire shall be photometered on a goniophotometer with absolute calibration. Its mounting orientation on the goniophotometer shall be identical to that used when it is applied to floodlight the screen. This luminaire shall be photometered in vertical and horizontal angular steps not exceeding 2.5°, over the angular range corresponding to the vertical and horizontal angular ranges subtended by the screen.

After the standard reference luminaire is calibrated, it shall be placed on the test stand to floodlight the

screen. Its reference photometric center used for positioning on the goniophotometer shall be located at $X = Y = Z = 0$, and its reference axis shall be aimed at the screen point $X = Z = 0$. It shall be operated at a current identical to that used for the goniophotometer test. The digital camera system shall be used to collect data for the entire screen area, providing a bit count for each pixel. Dark field readings shall be subtracted as describe in **Section 7.1**. An exposure time shall be used that produces bit count data within the usable signal range for the entire screen.

A calibration factor, F , for each pixel shall be determined:

$$F_{RC} = \frac{I'}{PV'_{RC}}, \text{ in candelas per bit,} \quad (8-1)$$

where:

R = row number for the pixel

C = column number for the pixel

I' = intensity received from the reference luminaire at the screen point corresponding to the pixel

PV'_{RC} = bit count or pixel value for that pixel, using the standard reference luminaire

The value of I' is extracted from the goniophotometer test data file, for the applicable horizontal and vertical angles of interest. Determination of these angles for a given pixel is described in **Annex A**.

The F value for each pixel may be stored in an electronic file. During subsequent photometry of a test luminaire, the F value for a given pixel is applied to the measured bit count for that pixel to determine the test luminaire intensity, I_A . (Note, however, that a correction for exposure time may be needed; see **Section 8.4**.)

$$I_A = PV_{RC} \cdot F_{RC}, \quad (8-2)$$

where PV_{RC} is the bit count or pixel value for a pixel in row R and column C , using the test luminaire

Note: **Equation 8-2** assumes that an identical exposure time is used for the calibration and the test. In practice, if different exposure times are used, a correction is required, (see **Section 8.4**).

From the I_A data, an absolute test report can be produced for the captured angular range.

Derivation of the test luminaire intensity is independent of the screen reflectance. While a diffuse screen is desirable to eliminate specular reflections, perfectly uniform reflectance is not required.

Individual pixels will vary in their sensitivity, and the value of F will be different for each pixel. The recommended calibration procedure eliminates the need for the photometrist to be concerned with such variables, as the value of F for all pixels is directly measured and stored.

8.4 Calibration Correction for Exposure Time

Equation 6-1 (see **Section 6.5**) is based on the use of an identical exposure time for the standard reference luminaire and the test luminaire. In practical cases, however, the test luminaire will generate different screen luminances from the reference luminaire, and a different exposure time may be necessary in order to ensure that the captured image of the test luminaire lies within the usable data range.

Camera evaluation, as described in **Section 7.3**, will have confirmed linear response versus exposure time. However, latent exposure time (see **Section 7.3** and **Figure 7-1**), needs to be accounted for in applying an exposure time correction factor for those cameras equipped with a shutter that exhibits latency.

The correction factor for exposure time, C_{ET} , is given by:

$$C_{ET} = \frac{T_R - L}{T_L - L}, \quad (8-3)$$

where:

T_R = Exposure time use for the reference standard

T_L = Exposure time used for the test luminaire

L = Latent exposure time derived from **Figure 7-1** (see **Section 7.2**)

If, for example, there is no latent exposure time ($L = 0$), and if T_R is 1 second and T_L is 2 seconds, the intensity values computed using **Equation 6-1** (see **Section 6.5**) for the test luminaire are multiplied by 0.50. If the latent exposure time (L) is 0.05 seconds, the correction factor is $(1 - 0.05)/(2 - 0.05) = 0.487$.

Applying the exposure-time correction factor modifies **Equation 6-1** thus:

$$I_A = PV_{RC} \cdot F_{RC} \cdot C_{ET}, \quad (8-4)$$

Equation 8-4 is used to determine absolute intensity values.

It should be noted that calibration correction for exposure time is unnecessary for systems that employ a fixed set of exposure times. In that case a calibration for each exposure time is needed.

8.5 Relative Intensity Calibration

It is often desirable to eliminate the variations in lamp output and prorate test results to manufacturer rated lamp lumens. In this case, a relative intensity calibration is needed.

Calibration and luminaire testing shall be conducted as described in Section 8.3. In addition, the bare-lamp luminous flux (lumens) shall be measured, for electrical conditions identical to those used for the digital photometry test. This may be achieved most simply using an integrating sphere.¹⁷⁻¹⁹

$$\text{lamp factor, } LF = \frac{\text{rated lamp lumens}}{\text{measured lamp lumens}}, \quad (8-5)$$

The rated lamp lumens shall be obtained from the manufacturer's catalog, or the average of several manufacturers shall be used.

The relative intensity, I_R , for each photometric reading, based on rated lamp lumens, shall be computed:

$$I_R = PV_{RC} \cdot F_{RC} \cdot LF \quad (8-6)$$

From the I_R data, a relative test report can be produced for the captured angular range.

9.0 Test Procedure

The DUT to be tested shall be prepared in accordance with appropriate recommended testing procedures.^{17-24,27} It shall be aligned with the screen such that a known reference axis of the DUT is perpendicular to the screen. DUT stabilization shall be per recommended procedures.^{17-24,27}

The offset distances between the camera and luminaire photometric center, X_{LC} , Y_{LC} and Z_{LC} , shall be noted.

A camera exposure shall be made using a nominal exposure time. If the maximum pixel reading exceeds the useable range as defined in **Section 7.2**, further exposures shall be made using reduced exposure times to obtain a maximum reading within the useable range. Photometer systems that utilize a system of different exposure times, integrating the useable data into a single array of values described in **Section 6.5** are exempt from this process. It is recommended that if the maximum pixel reading lies at less than 80% of the highest bit count that defines the useable range, a further exposure should be made, using an increased exposure time, calculated to obtain a maximum reading between 80 and 100% of the useable range limit. Note that per **Section 7.2** the usable range upper limit is defined by system linearity and will be less than the maximum limit of the camera.

A dark field exposure shall be captured using the same exposure time as the final exposure used for the test luminaire, unless dark field levels have been shown to be negligible.

Readings shall be corrected for dark field by subtracting the dark field value for each pixel from the corresponding test exposure values.

10.0 Data Reduction and Test Report

An array of intensity data shall be constructed for the vertical and horizontal angular range of interest, in the desired test increments. The method for construction of this array is as follows:

1. For each pixel, determine the corresponding intensity, either relative or absolute (see **Section 8**).
2. For each pair of angles that constitutes a desired data point, compute the pixel array row and column counters, R and C , per **Annex A**, for the desired coordinate system. (Note that the calculated values of R and C are unlikely to be integers; i.e., the point

on the CCD array corresponding to the desired screen test point is likely to lie between pixels.)

3. Compute the intensity for the desired point using bilinear interpolation of the measured intensity data for the four pixels surrounding the exact pixel array point, using the known vertical and horizontal resolution of the system. This will yield accurate interpolation, provided the angular resolution requirements of **Section 3.0** are followed.

Given absolute or relative intensity data for the desired array of test angles, a test report may be produced in accordance with applicable IES procedures.²⁰⁻²⁴ Reduction of the intensity data is identical to that for data collected using a goniophotometer, although the vertical and horizontal angular range may be less.

An electronic data file may be produced per IES procedures.^{25,26} Note, however, that the test system may not fulfill the angular range requirements of References 25 and 26, as a full hemisphere of data cannot be collected unless a dual axis rotating stage is used. The data file shall conform to the starting and ending vertical and horizontal angle requirements of Reference 25 or 26. An intensity of zero shall appear for any directions that are not measured.

The measured angular range, vertical and horizontal, shall be indicated in the test report. It should be noted that due to the limited angular range of the measurements, full test reports are not possible for all classes of luminaires.

Glossary

A/D converter: Analog-to-digital converter. Electronic circuitry that converts an analog voltage or current into a digital number proportional to the magnitude of the voltage or current.

ADU: Analog-to-digital unit. Also referred to as a *bit*. The relationship between the ADUs generated and the number of electrons acquired on the CCD is defined by the system gain.

bias: In a CCD camera system, the minimum intensity required for each exposure (equivalent to performing a zero-second exposure with the shutter closed). Without adding any light, the bias allows charge to be read out on the CCD while raising the intensity level high enough to ensure that the camera does not deliver a negative number to the A/D converter. (The A/D converter only works in the set of positive numbers and has no instructions for processing negative numbers.) The bias, which is not user selectable, is set at the factory and remains stable over the lifetime of the camera system.

blooming: The charge spill-over that occurs with a CCD pixel reaches its full well capacity.

CCD: Charge-coupled device. A light-sensitive silicon chip often used as a photodetector in digital camera systems. CCDs are manufactured in a wide variety of formats, architectures, and grades. The charge in CCD pixels is measured sequentially and the measurement process empties the pixel of charge.

charge: In CCD imaging technology, a measure of the number of electrons.

CID: Charge-injection device. Similar to a CCD except that the charge in individual pixels can be measured non-destructively at random.

CMOS: Complementary metal oxide semiconductor. Semiconductor technology used to manufacture electronic devices. Also used to manufacture light-sensitive devices similar to CCDs.

cooled CCD: A charge-coupled device that is operated below ambient temperature in order to reduce or eliminate dark current. CCD cooling is often achieved via Peltier (thermoelectric) coolers.

dark current: The charge accumulated within a well, in the absence of light. Cooling the CCD can reduce dark current. Also called *thermally generated charge*.

dark noise: The statistical variation of the dark current equal to the square root of the dark current. Dark current can be subtracted from an image, while dark noise remains. Also called *dark current noise*.

digitize: To convert data (or images) into a digital format.

dynamic range: The ratio of the maximum (brightest) to minimum (darkest) signal levels that can be detected by a camera.

e/ADU: See *gain*.

exposure time: The length of time that a CCD is accumulating charge.

frame: An entire data set from a CCD array.

full-frame CCD: The simplest type of CCD. Full-frame CCDs use the entire active pixel array to expose photons and to integrate and transport charge. They utilize a shutter to control the exposure and block light during CCD readout, preventing charge smearing.

full well capacity: The number of electrons that can be held in one potential well. Charge generated in excess of the full well capacity can cause blooming.

gain: In digital cameras, gain defines the relationship between the number of electrons acquired on a CCD and the analog-to-digital units (ADUs) generated.

image averaging: A method of reducing random image noise by averaging a pixel's digital readout throughout a series of successive frames.

interline-transfer CCD: Also called *interline CCD* or *progressive-scan CCD*. This type of CCD chip has masked areas on alternate columns of the sensor for storage. Once an exposure is completed, the charge in the light sensitive pixels is transferred quickly to the pixels in the adjacent masked column.

Lambertian reflector: A surface that exhibits Lambertian reflectance scatters light falling on it such that the apparent brightness of the surface to an observer is the same regardless of the observer's direction of view.

latency: The delay that occurs between the instant of issuing the command to start an exposure and the

instant at which the shutter actually opens. This is observed in some mechanically shuttered systems.

linearity: In CCD imaging technology, precise linearity dictates that an object that is twice as luminous as another object will be digitized to exactly twice the number of ADUs (bits) as in the resultant image after all bias and dark corrections have been made.

noise: An unwanted or undesirable signal. In a CCD camera system, internally generated interference or a number of other factors can cause unwanted signal to appear in an image. Many noise sources exist, but three sources account for the majority of total noise: dark noise, photon (shot) noise, and read noise.

output amplifier: Electronics in the CCD that amplify the electrons sufficiently to get the signal to the A/D converter. The output amplifier is the primary source of read noise. Also called a *readout amplifier*.

Peltier effect: The transfer of heat in the opposite direction of a current flow. By pumping current through a "Peltier stack," heat can be caused to flow from a CCD chip to a heat sink.

photon (shot) noise: Noise that is a fundamental property of the quantum nature of light. Photon noise is always present in an imaging system.

pixel: Short for *picture element*. The smallest element in a visual display or smallest light sensitive element in a CCD array.

Poisson distribution: A probability function used to model the density of counts of a randomly occurring event obtained during a specified interval.

potential well: In a CCD, a discrete region within the device's imaging array where an electron resulting from an incident photon may be trapped.

read noise: In CCD imaging technology, unwanted signal that is generated by the on-chip output amplifier.

resolution: A measure of how fine of a detail can be detected, in terms of space (spatial resolution), angle (angular resolution), or intensity.

smear: Residual charge left behind in potential when an image is shifted within a CCD. Occurs when the image is shifted while the light is falling on the sensor.

SNR: Signal-to-noise ratio. In CCD imaging technology, the measure of the signal quality at a given pixel, expressed as the ratio of the measured signal to the overall noise at that pixel.

thermoelectric cooling: The process of removing heat from a CCD chip by using solid-state Peltier cooling devices.

Annex A – Geometrical Relationships

A useable screen-photometer test system employing a digital camera requires that:

- Geometric compensation be made for the distance of separation between the camera and the luminaire
- Methodology be available to relate a given CCD pixel to a particular point on the screen
- A given point on the screen be identifiable in terms of the vertical and horizontal photometric angles in reference to the luminaire
- These relationships be expressed for various photometric coordinate systems

The above geometrical relationships are derived, and each pixel reading (bit count) can then be identified with the vertical and horizontal photometric angles of the corresponding screen point. Measurement of the entire pixel array corresponding to the screen area then can be used to yield the photometric report, after proper calibration.

A.1 Pixel Rows and Columns

Each pixel in a row of pixels in the CCD array is denoted by a value C , or column number. Each pixel in a column of pixels is denoted by a value of R , or row number. Thus, any pixel can be designated by R, C .

Figure A-1 illustrates the screen and a diagram of the CCD as shown from behind the CCD, viewing toward the screen. The screen has a crosshairs superimposed, centered at the reference center point of the screen, where $X = Z = 0$, and $Y = D$ (see **Section 3.0**).

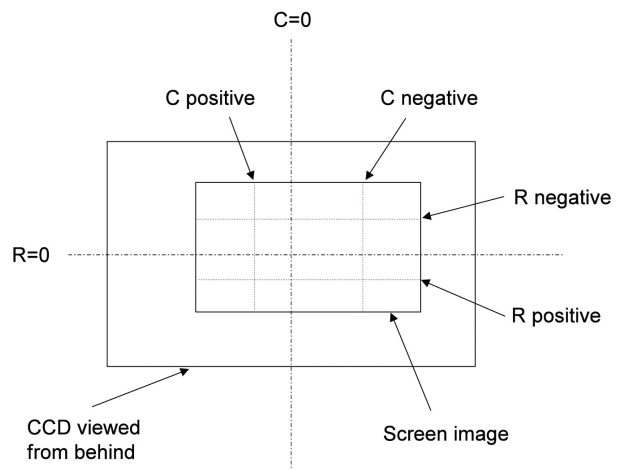


Figure A-1. Definition of CCD Row and Column Counters, R and C. (© Illuminating Engineering Society)

If the camera is located above the luminaire (Z_{LC} is positive; see **Section 3.0**), and to the right of the luminaire when viewed from behind looking towards the screen, (X_{LC} is positive), then the image of the screen on the CCD array will be off-center, shifted upward and toward the right due to inversion of the image.

If $R = 0$ for the row of pixels corresponding to the horizontal crosshairs line, and $C = 0$ for the column of pixels corresponding to the vertical crosshairs line, then the crosshairs crossover point, which corresponds to screen optical center reference location ($X = Z = 0$, $Y = D$), is the origin of the pixel array as denoted by values of R and C .

Points on the screen above the $Z = 0$ line are considered positive ($+Z$), and corresponding rows of pixels below the $R = 0$ row have positive values of R (see **Figure A-1**). Rows of pixels above the $R = 0$ row have negative values of R .

Points on the screen to the right of the $X = 0$ line are considered positive ($+X$), and corresponding columns of pixels left of the $C = 0$ column are positive (see **Figure A-1**). Columns of pixels to the right of the $C = 0$ line have negative values of R .

A.2 Relating Screen Locations and Photometric Angles – Type B Coordinates

This section derives the relationship between the horizontal and vertical photometric angles, designated HP_B and VP_B , and the corresponding screen coordinates, X and Z .

This derivation is performed for a type B coordinate system, designated by suffix B for the photometric angles.¹⁶ (See **Figure A-2**.)

O is the origin, depicting the reference center of the screen.

L is the photometric center of the luminaire.

S is a point on the screen corresponding to photometric angles HP_B and VP_B having coordinates X_S and Z_S .

The loci of constant horizontal photometric angle, HP_B , and constant vertical photometric angle, VP_B , are shown, resulting from the projection of a type B coordinate photometric web onto a flat screen.

$$VP_B = \tan^{-1}\left(\frac{AO}{D}\right)$$

and

$$Z_S = AO$$

Therefore:

$$VP_B = \tan^{-1}\left(\frac{Z_S}{D}\right) \quad (\text{A-1})$$

Further:

$$HP_B = \tan^{-1}\left(\frac{AS}{AL}\right)$$

$$\cos(VP_B) = \frac{OL}{AL} = \frac{D}{AL}$$

Therefore:

$$HP_B = \tan^{-1}\left[\frac{(AS)\cos(VP_B)}{D}\right]$$

Further:

$$X_S = AS$$

Therefore:

$$HP_B = \tan^{-1}\left[\frac{X_S \cos(VP_B)}{D}\right] \quad (\text{A-2})$$

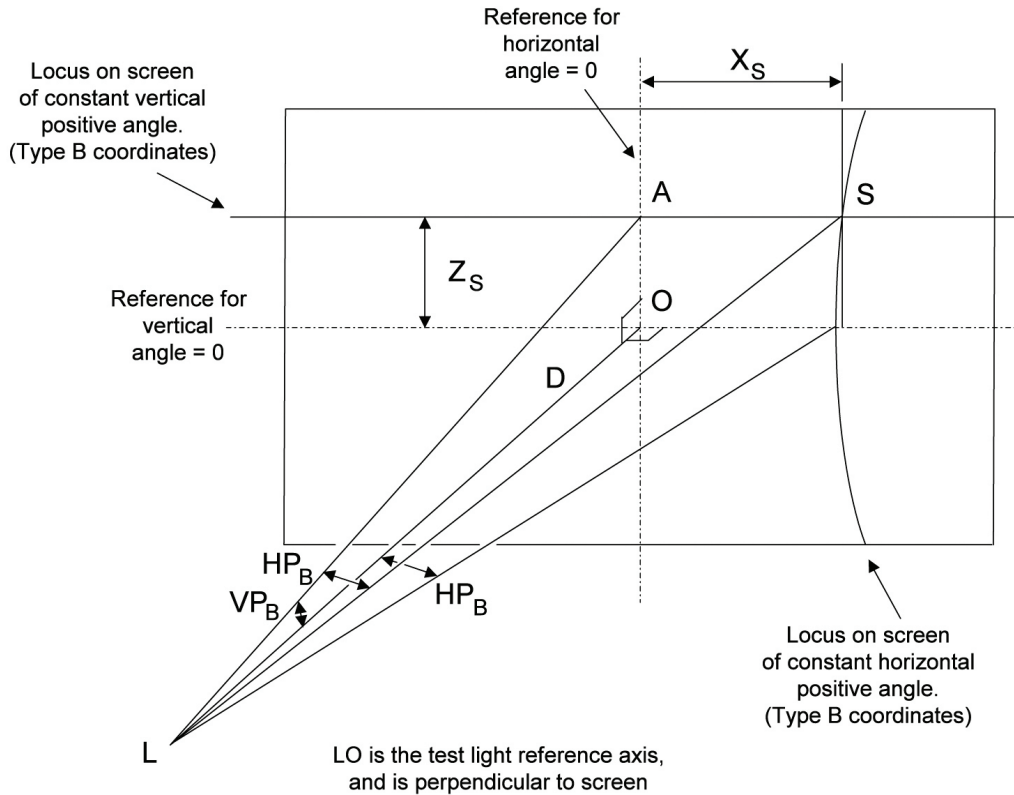


Figure A-2. Type B coordinate light projection onto screen. (© Illuminating Engineering Society)

Thus, given the coordinates of any screen location, the corresponding vertical and horizontal photometric angles can be determined from **Equations A-1** and **A-2**, for a type B coordinate system.

A.3 Relating Screen Geometry to Pixel Row and Column (Type B Coordinates)

Sections A.3.1 and **A.3.2** allow the pixel row and column to be determined for given photometric angles. **Section A.3.3** provides the converse, allowing photometric angles to be determined for a given pixel.

In the preceding section, photometric angles have been related to screen geometry for the type B coordinate system. By relating pixel locations in the CCD array, as defined by row and column counters described in **Section A.1** to the equivalent screen coordinates, photometric vertical and horizontal angles can then be related to a given pixel location.

A.3.1 Determining the Pixel Row for a Given Vertical Photometric Angle (Type B Coordinates). **Figure A-3** illustrates an arrangement of screen, camera and luminaire in vertical section, equivalent to **Figure A-2** (see **Section A.2**), for a type B coordinate system.

The camera is offset from the luminaire by distances Y_{LC} and Z_{LC} , both positive.

If VC_B = the vertical angle in type B coordinates from the camera reference point to the point of interest on the screen, then:

$$VC_B = \tan^{-1} \left(\frac{Z_S - Z_{LC}}{D - Y_{LC}} \right)$$

and:

$$\tan VP_B = \frac{Z_S}{D}$$

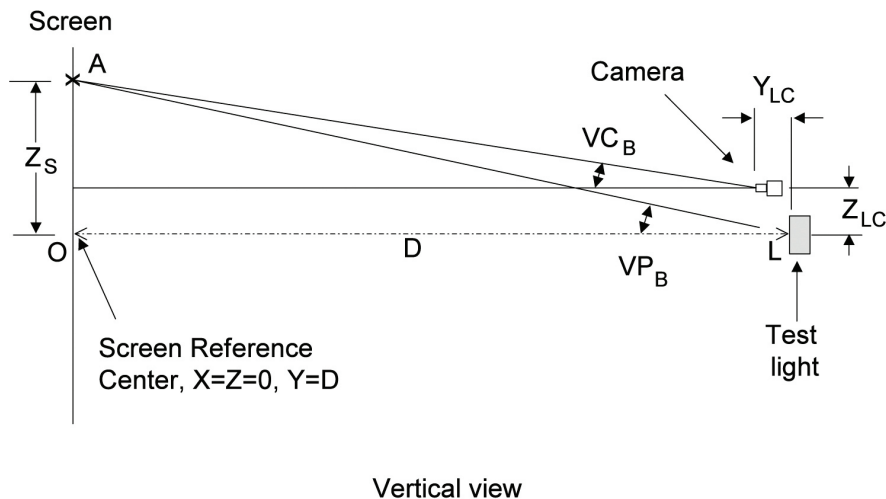
Therefore:

$$VC_B = \tan^{-1} \left[\frac{D \tan(VP_B) - Z_{LC}}{D - Y_{LC}} \right] \quad (\text{A-3})$$

for a type B coordinate system.

The above method provides the vertical angle from the camera reference point to the point of interest on the screen, incorporating the transposition distances, Y_{LC} and Z_{LC} .

Assuming the image has been corrected for lens distortions, the principles of geometric optics can be used to determine the value of the pixel row counter R corresponding to the vertical photometric angle VP_B . From geometric optics, it is known that for a given lens



LO is the test light reference axis,
and is perpendicular to screen

Figure A-3. Vertical coordinates of test light, camera, and screen through vertical reference axis. (© Illuminating Engineering Society)

of focal length f , the distance d of a point on the focal plane is related to VC_B by:

$$d = f \tan VC_B$$

Therefore:

$$d = f \left[\frac{D \tan(VP_B) - Z_{LC}}{D - Y_{LC}} \right] \quad (A-4)$$

The pixel row counter R , can be determined from the pixel height, h :

$$R = \frac{d}{h}$$

Substitution gives:

$$R = \left(\frac{f}{h} \right) \left[\frac{D \tan(VP_B) - Z_{LC}}{D - Y_{LC}} \right] \quad (A-5)$$

for a type B coordinate system.

The values of D , Y_{LC} and Z_{LC} are fixed and measured distances. The values of f and L are known from lens and camera specifications. Therefore, for any row of pixels as denoted by the row counter R , the vertical photometric angle in type B coordinates, VP_B , can be determined.

A.3.2 Determining the Pixel Column for Given Photometric Angles (Type B Coordinates). Figure A-4 provides a horizontal view of the screen, camera and luminaire arrangement, equivalent to Figure A-2 (see Section A.2).

Although it is required to determine the pixel column counter C from B type photometric coordinates, the calculations are simplified if the A type photometric coordinate system is used initially. HC_A = the horizontal angle in type A coordinates from the camera reference point to the point of interest, S , on the screen (see Figure A-5), and VP_B and HP_B are the vertical and horizontal photometric angles, respectively, in type B coordinates.

Then:

$$HC_A = \tan^{-1} \left(\frac{BE}{BF} \right) = \tan^{-1} \left(\frac{X_S - X_{LC}}{D - Y_{LC}} \right) \quad (A-6)$$

and:

$$(\tan HP_B) = \frac{X_S}{AL} \quad (A-7)$$

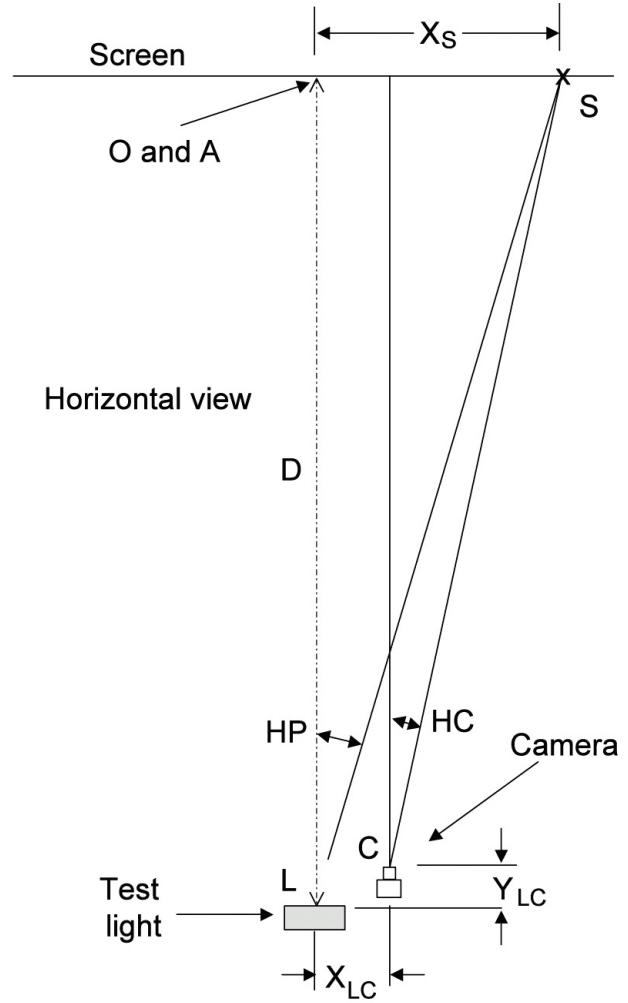


Figure A-4. Horizontal coordinates of test light, camera, and screen through horizontal reference axis.

(© Illuminating Engineering Society)

further:

$$\cos VP_B = \frac{D}{AL}$$

Substituting for AL in Equation A-7:

$$\tan(HP_B) = \frac{X_S \cos(VP_B)}{D}$$

Therefore:

$$X_S = \frac{D \tan(HP_B)}{\cos(VP_B)}$$

Substituting in Equation A-6:

$$HC_A = \tan^{-1} \left\{ \left(\frac{1}{D - Y_{LC}} \right) \left[\frac{D \tan(HP_B)}{\cos(VP_B)} - X_{LC} \right] \right\} \quad (A-8)$$

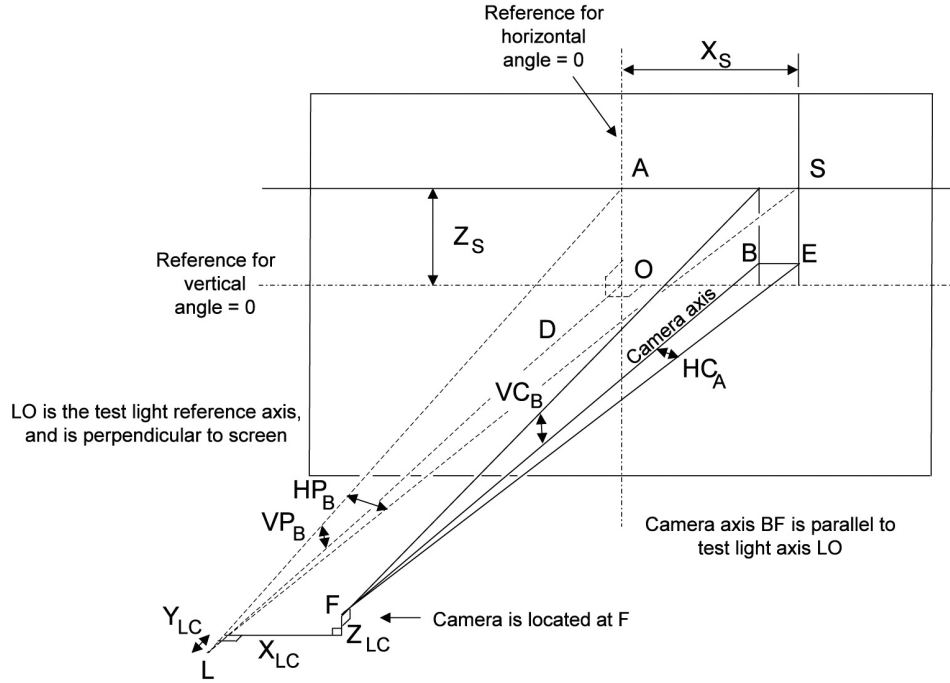


Figure A-5. Type B coordinate light projection and camera location. (© Illuminating Engineering Society)

From geometric optics:

$$\tan HC_A = \frac{d}{f}$$

Solving for d and substituting in **Equation A-8**:

$$d = \left(\frac{f}{D - Y_{LC}} \right) \left[\frac{D \tan(HP_B)}{\cos(VP_B)} - X_{LC} \right] \quad (\text{A-9})$$

The pixel column counter C can be determined from the pixel width w :

$$C = \frac{d}{w}$$

Substitute in **Equation A-9**:

$$C = \left[\frac{f}{w(D - Y_{LC})} \right] \left[\frac{D \tan(HP_B)}{\cos(VP_B)} - X_{LC} \right] \quad (\text{A-10})$$

The values of D , Y_{LC} , and X_{LC} are fixed and measured distances. The values of w and f are known from camera and lens specifications. Therefore, for any column of pixels denoted by the column counter C , the horizontal photometric angle in type B coordinates, HP_B , can be determined.

It is important to note that the value of C , the pixel column number for a given photometric horizontal angle in type B coordinates, will change as the vertical

photometric angle changes. This is because a type B coordinate system projects a curved line onto a flat screen for a constant horizontal angle (see **Figure A-2**, in **Section A.2**). Thus, a given pixel column covers a range of horizontal angles.

A.3.3 Determining Photometric Angles for a Given Pixel (Type B Coordinates).

The equations derived in the previous sections may be revised and used to determine the vertical and horizontal photometric angles for any given pixel, as specified by its row and column counters.

A.3.3.1 Vertical Photometric Angle. Equation A-5 (see **Section A.3.1**) may be rewritten in terms of VP_B , allowing the determination of the vertical photometric angle corresponding to a given row of pixels denoted by the row counter R .

$$VP_B = \tan^{-1} \left[\frac{(D - Y_{LC})(hR/f) + Z_{LC}}{D} \right] \quad (\text{A-11})$$

for a type B coordinate system.

The values of h and f are known, as are the camera offset distances; thus, VP_B is known.

It is important to note that the vertical photometric angle is constant for given pixel row, as long as a type B coordinate system is in use.

A.3.3.2 Horizontal Photometric Angle. Equation A-9 can be rewritten:

$$HP_B = \tan^{-1} \left\{ \left[\frac{\cos(VP_B)}{D} \right] \left[\frac{d(D - Y_{LC})}{f} + X_{LC} \right] \right\}$$

From geometric optics, $C = d/w$. Substituting for d :

$$HP_B = \tan^{-1} \left\{ \left[\frac{\cos(VP_B)}{D} \right] \left[\frac{wC(D - Y_{LC})}{f} + X_{LC} \right] \right\} \quad (\text{A-12})$$

for a type B coordinate system.

The values of VP_B , w , and f are known, as are the camera offset distances; thus, HP_B is known.

Using the equations in this section and in **Section A.3.3.1**, the pair of type B photometric angles can be determined for any given pixel, as denoted by its row and column counters R and C .

A.4 Use of a Type A Coordinate System

In **Section A.3**, the geometric relationship between photometric angles and pixel locations was derived for a type B coordinate system. To work with a type A coordinate system, shown in **Figure A-6**, conversion formulae can be used to convert between type A and type B coordinates. The equations of **Section A.3** can subsequently be applied.

A.4.1 Relating Screen Locations and Photometric Angles (Type A Coordinates). VP_A and HP_A are the vertical and horizontal photometric angles in a type A coordinate system, equivalent to angles VP_B and HP_B in the type B coordinate system, as covered in **Section A.3**.

For a chosen screen location, the applicable type B photometric angles, VP_B and HP_B , are computed using **Equations A-1** and **A-2** (see **Section A.2**). HP_A can then be determined.

From **Figure A-6**:

$$\tan(HP_A) = \frac{X_S}{D}$$

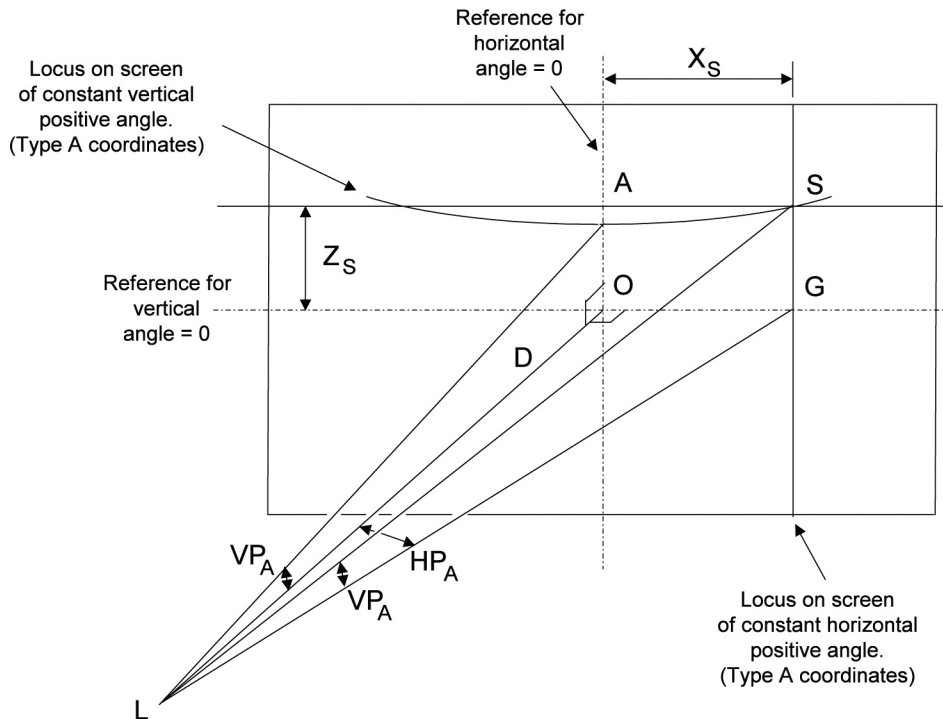


Figure A-6. Type A coordinate light projection onto screen. (© Illuminating Engineering Society)

From **Figure A-2**:

$$\tan(HP_B) = \frac{X_S}{AL}$$

Therefore:

$$\tan HP_A = \frac{AL \tan(HP_B)}{D} \quad (\text{A-13})$$

From **Figure A-2**:

$$\cos(VP_B) = \frac{D}{AL}$$

Substituting for AL in **Equation A-13**:

$$HP_A = \tan^{-1} \left[\frac{\tan(HP_B)}{\cos(VP_B)} \right] \quad (\text{A-14})$$

From **Figure A-6**:

$$\sin(VP_A) = \frac{Z_S}{LS}$$

From **Figure A-2**:

$$\cos(HP_B) = \frac{AL}{LS}$$

Therefore:

$$\sin(VP_A) = \frac{Z_S \cos(HP_B)}{AL} \quad (\text{A-15})$$

From **Figure A-2**:

$$\sin(VP_B) = \frac{Z_S}{AL}$$

Substituting for AL in **Equation A-15**:

$$VP_A = \sin^{-1} [\sin(VP_B) \cos(HP_B)] \quad (\text{A-16})$$

A.4.2 Relating Screen Geometry to Pixel Row and Column (Type A Coordinates). The pixel row and column counters, R and C , related to a pair of photometric angles, VC_B and HC_B , in the type B coordinate system can be determined using **Equations A-5, A-9, and A-10** (see **Section A.3**). To determine the R and C values for a pair of type A coordinate angles, the type A angles are first converted into type B angles using **Equations A-16 and A-17** (below). **Equations A-5, A-9, and A-10** are then used to find the pixel row and column counters, R and C .

From **Figure A-6**:

$$\tan(VP_A) = \frac{Z_S}{GL}$$

and:

$$\cos(HP_A) = \frac{D}{GL}$$

Therefore:

$$\frac{\tan(VP_A)}{\cos(HP_A)} = \frac{Z_S}{D}$$

Substituting in **Equation A-1**:

$$VP_B = \tan^{-1} \left[\frac{\tan(VP_A)}{\cos(HP_A)} \right] \quad (\text{A-17})$$

From **Figure A-2**:

$$\sin(HP_B) = \frac{X_S}{LS} \quad (\text{A-18})$$

From **Figure A-6**:

$$X_S = GL \sin(HP_A)$$

From **Figure A-6**:

$$LS = \frac{GL}{\cos(VP_A)}$$

Substituting for X_S and AL in **Equation A-18**:

$$VP_A = \sin^{-1} [\sin(HP_A) \cos(VP_A)] \quad (\text{A-19})$$

A.4.3 Determining Photometric angles for a Given Pixel (Type A Coordinates). To find the pair of angles, VP_A and HP_A , in a type A coordinate system related to a certain pixel, the related angles in a type B coordinate system, VP_B and HP_B , are first determined, using the methodology **Equations A-11 and A-13** (see **Section A.3.3**). Then **Equations A-14 and A-15** are used to find the type A equivalent angles, VP_A and HP_A .

A.5 Use of a Type C Coordinate System

Section A.3 has derived the geometric relationship between photometric angles and pixel locations, for a type B coordinate system. To work with a type C coordinate system, **Figure A-7**, conversion formulae can be used to convert between C and B coordinates, then subsequently the equations of **Section A.3** can be applied.

A.5.1 Relating Screen Locations and Photometric Angles (Type C Coordinates). VP_C and HP_C are the vertical and horizontal photometric angles in a type C coordinate system, equivalent to angles VP_B and HP_B in the type B coordinate system, as covered in **Section A.3** (see **Figure A-7**).

**Approved Method: Intensity Distribution Measurement of Luminaires
and Lamps Using Digital Screen Imaging Photometry**

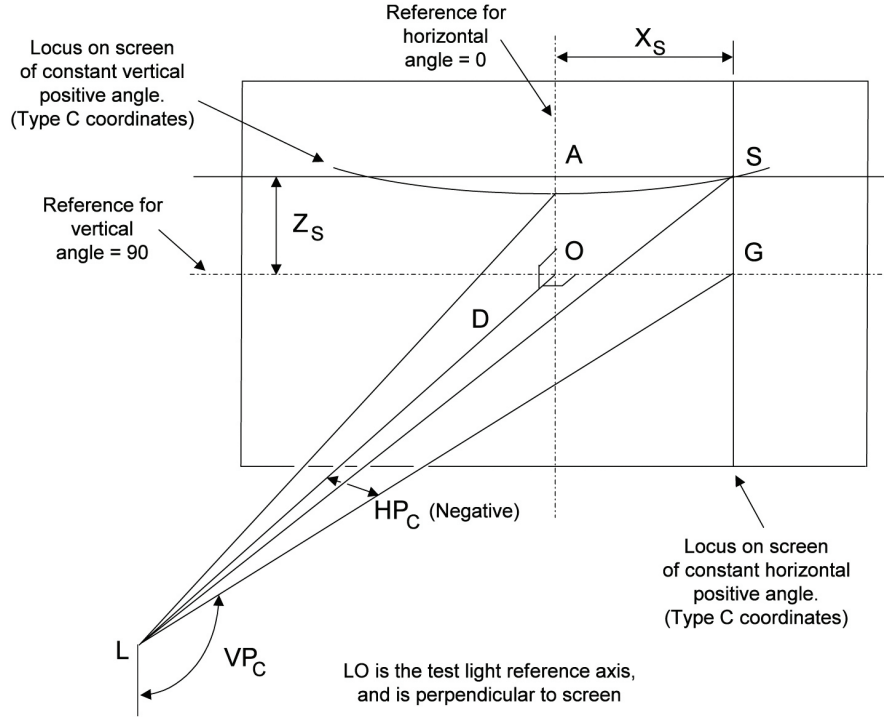


Figure A-7. Type C coordinate light projection onto screen. (© Illuminating Engineering Society)

For a chosen screen location, the applicable type B photometric angles, VP_B and HP_B , are computed using **Equations A-1** and **A-2**. Horizontal angles for type C coordinates are identical, but negative, to those for type A.

From **Equation A-14**:

$$HP_C = -\tan^{-1} \left[\frac{\tan(HP_B)}{\cos(VP_B)} \right] \quad (\text{A-20})$$

Type C vertical angles are identical to those for type B coordinates, but are increased by 90° . (That is, the horizontal plane through the light center has a vertical angle of 90° for type C but 0° for type B coordinates.)

From **Equation A-16**:

$$VP_C = \sin^{-1} [\sin(VP_B) \cos(HP_B) + 90] \quad (\text{A-21})$$

A.5.2 Relating Screen Geometry to Pixel Row and Column (Type C Coordinates). The pixel row and column counters, R and C , related to a pair of photometric angles, VC_B and HC_B , in the type B coordinate system can be determined using **Equations A-5**, **A-9**, and **A-10** (see **Section A.3**). To determine the R and C values for a pair of type C coordinate angles,

the type C angles are first converted into type B angles using **Equations A-20** and **A-21**, below. **Equations A-5**, **A-9**, and **A-10** are then used to find the pixel row and column counters, R and C .

Using the same logic as **Section A.5.1**, **Equation A-17** can be modified to find VP_B from type C coordinate angles. The value for $\cos(HP_C)$ remains positive when HP_C is negative.

$$VP_B = \tan^{-1} \left[\frac{\tan(VP_C - 90)}{\cos(HP_C)} \right] \quad (\text{A-22})$$

Equation A-19 can be modified to find HP_B from type C coordinate angles.

$$\cos(VP_A) = \sin(VP_C)$$

Therefore:

$$HP_B = -\sin^{-1} [\sin(HP_C) \sin(VP_C)] \quad (\text{A-23})$$

A.5.3 Determining Photometric Angles for a Given Pixel (Type C Coordinates). To find the pair of angles, VP_C and HP_C , in a type C coordinate system related to a certain pixel, the related angles in a type B coordinate system, VP_B and HP_B , are first determined, using **Equations A-11** and **A-12** (see **Section A.3.3**).

Equations A-20 and A-21 are then used to find the type C equivalent angles, VP_C and HP_C .

A.6 Use of a Modified Type C Coordinate System

Goniophotometric measurements using a type C coordinate system frequently involve aiming a luminaire at nadir. The angular web, as projected on the floor of the photometer chamber will therefore be formed from circles, each representing a vertical angle, and radial lines, each representing a horizontal angle. If a lamp is aimed horizontally at a screen and this photometric web is applied about its aiming axis, the web is depicted on the screen as circles and radial lines. **Figure A-8**. This will be referred to as a *modified type C coordinate system*; it is a type C coordinate system tilted 90 degrees vertically, with the polar axis of the coordinate system aimed perpendicular to a vertical screen.

Under certain conditions, it may be desirable to conduct digital photographic photometry using this modified coordinate system. The geometric relationships are described below.

The terms *horizontal* and *vertical* as angle descriptors are not strictly applicable for the modified type C coordinate system, but will be used so as to be equivalent to the type A, B, and C systems described in previous sections. The meanings of *horizontal angles* and *vertical angles* for the modified type C system are clarified in **Figure A-8**.

The “horizontal” angle planes, as described by radial lines from the pole, use the positive Z axis on the screen as the zero-degree angle.

The geometric relationship between photometric angles and pixel locations, for a type B coordinate system was derived in **Section A.3**. To work with the modified type C coordinate system, shown in **Figure A-8**, conversion formulae can be used to convert between type C modified coordinates and type B coordinates, and then the equations of **Section A.3** can be applied.

A.6.1 Relating Screen Locations and Photo-metric Angles (Modified Type C Coordinates). VP_{CM} and HP_{CM} are the vertical and horizontal photometric angles in a modified type C coordinate system equivalent to angles VP_B and HP_B in the type B coordinate system, as covered in **Section A.3**.

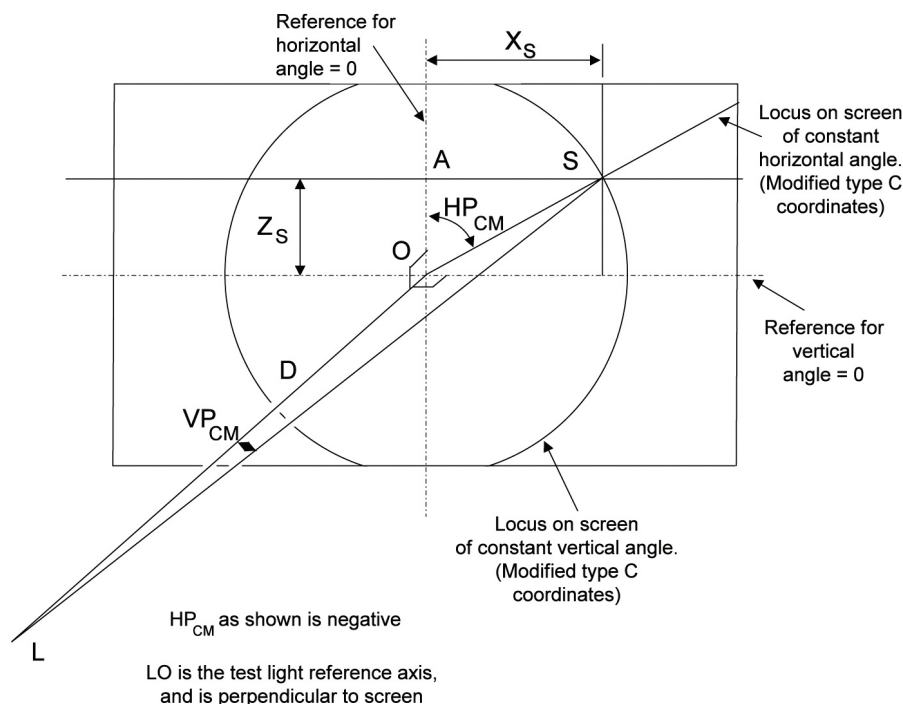


Figure A-8. Modified type C coordinate light projection onto screen. (© Illuminating Engineering Society)

For a chosen screen location, the applicable type B photometric angles, VP_B and HP_B , are computed using **Equations A-1** and **A-2** (see **Section A.2**). HP_{CM} may be determined from **Equation A-25**, below.

From **Figure A-2** (see **Section A.2**):

$$\tan(HP_B) = \frac{X_S}{AL}$$

and:

$$\sin(VP_B) = \frac{Z_S}{AL}$$

Therefore:

$$\tan(HP_B)\sin(VP_B) = \frac{X_S}{Z_S} \quad (\text{A-24})$$

From **Figure A-8**:

$$\tan(HP_{CM}) = \frac{-X_S}{Z_S}$$

Substituting in **Equation A-2**:

$$HP_{CM} = -\tan^{-1}\left[\frac{\tan(HP_B)}{\sin(VP_B)}\right] \quad (\text{A-25})$$

VP_{CM} may be determined from **Equation A-27**, below.

From **Figure A-2**:

$$\cos(HP_B) = \frac{AL}{LS}$$

and:

$$\cos(VP_B) = \frac{D}{AL}$$

Therefore:

$$\cos(HP_B)\cos(VP_B) = \frac{D}{LS} \quad (\text{A-26})$$

From **Figure A-8**:

$$\cos(VP_{CM}) = \frac{D}{LS}$$

Substituting in **Equation A-26**:

$$VP_{CM} = \cos^{-1}[\cos(HP_B)\cos(VP_B)] \quad (\text{A-27})$$

A.6.2 Relating Screen Geometry to Pixel Row and Column (Modified Type C Coordinates). The pixel row and column counters, R and C , related to a pair of photometric angles, VC_B and HC_B , in the type B coordinate system can be determined using **Equations A-5**, **A-9**, and **A-10** (see **Section A.3**). To determine the

R and C values for a pair of modified type C coordinate angles, the modified type C angles are first converted into type B angles using **Equations A-28** and **A-30**, below. **Equations A-5**, **A-9**, and **A-10** are then used to find the pixel row and column counters, R and C .

From **Figure A-8**:

$$\cos(HP_{CM}) = \frac{Z_S}{OS}$$

and:

$$\tan(VP_{CM}) = \frac{OS}{D}$$

Therefore:

$$\cos(HP_{CM})\tan(VP_{CM}) = \frac{Z_S}{D}$$

Substituting into **Equation A-1**:

$$VP_B = \tan^{-1}[\cos(HP_{CM})\tan(VP_{CM})] \quad (\text{A-28})$$

From **Figure A-8**:

$$\sin(HP_{CM}) = \frac{-X_S}{OS}$$

and:

$$\sin(VP_{CM}) = \frac{OS}{LS}$$

Therefore:

$$\sin(HP_{CM})\sin(VP_{CM}) = \frac{-X_S}{LS} \quad (\text{A-29})$$

From **Figure A-2**:

$$\sin(HP_B) = \frac{X_S}{LS}$$

Substituting in **Equation A-29**:

$$HP_B = -\sin^{-1}[\sin(HP_{CM})\sin(VP_{CM})]$$

A.6.3 Determining Photometric Angles for a Given Pixel (Modified Type C Coordinates). To find the pair of angles, VP_{CM} and HP_{CM} , in a modified type C coordinate system related to a certain pixel, the related angles in a type B coordinate system, VP_B and HP_B , are first determined using **Equations A-11** and **A-13** (see **Section A.3.3**). **Equations A-22** and **A-23** are then used to find the modified C equivalent angles, VP_{CM} and HP_{CM} .

A.7 Determination of Resolution

The resolution of the photometric angles provided by a camera system is dependent upon the location of the pixel being considered. As pixel width is constant over the CCD, a pixel at the outer edge will subtend a smaller angular range on the screen than a pixel at the center will. Resolution is therefore poorest for centrally located pixels. Thus, to compute minimum system resolution, resolution is determined for a centrally located pixel.

From **Figure A-9**, a central pixel of width w will subtend an angle A_H , where A_H is the photometric angular resolution, measured horizontally.

The value of w is computed using manufacturer data:

$$w = \frac{\text{width of CCD}}{\text{No. of pixel columns}}$$

A_H then is determined using geometric optics (see **Figure A-9**):

$$A_H = \tan^{-1}\left(\frac{w}{f}\right),$$

where f is the lens focal length.

The pixel height, h , is found from:

$$h = \frac{\text{height of CCD}}{\text{No. of pixel rows}}$$

If $h = w$, the vertical and horizontal resolutions are identical. Otherwise, the photometric angular resolution, measured vertically, A_V , is given by:

$$A_V = \tan^{-1}\left(\frac{h}{f}\right)$$

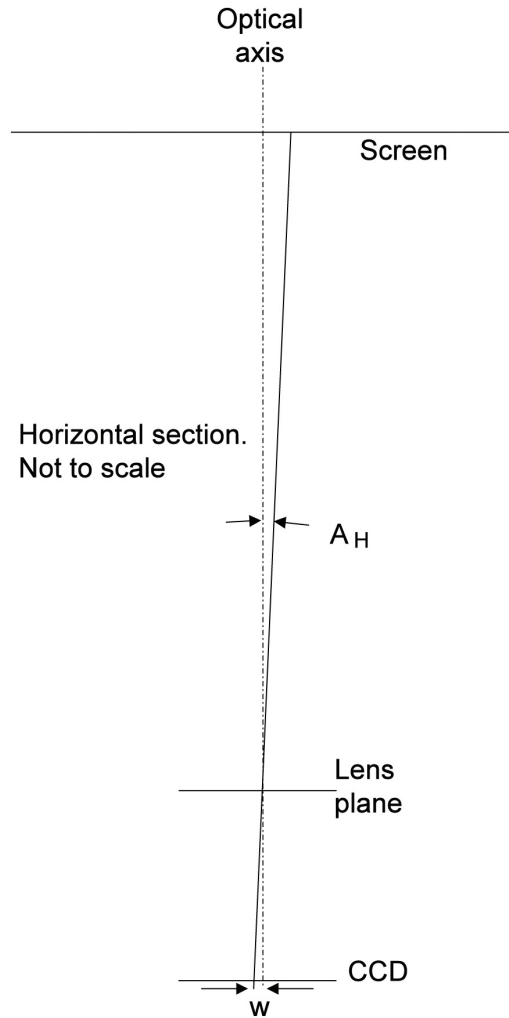


Figure A-9. Pixel and screen relationship. (© Illuminating Engineering Society)

Annex B – Lens Distortion Correction

A wide field of view is often desirable when making photometric measurements. However, the short-focal-length lenses that provide a wide field of view often exhibit barrel distortion, pincushion distortion, or a combination. Distortions of the image have to be corrected in order to accurately determine and use the photometric angles of the pixels. It is important to note that if the camera is pointed at the screen center rather than normal to the screen, an additional distortion correction will be required to remove the keystone effect.

B.1 Basic Principles

It is assumed that the lens distortions are axially symmetric. The axis of symmetry is the optical axis of the lens used to form images on the CCD array. If the lens optical axis is not perpendicular to the plane of the detector array, non-axially symmetric distortions can be produced, but such distortions are not addressed in this document.

The radial distortion correction is made by mapping the raw photometric data into a new image with the use of a polynomial transformation. The coordinates of the physical location of the pixels on the surface of the detector array are converted from Cartesian (x, y) coordinates to a polar coordinate system (R, θ), with the center pixel at the origin of the polar coordinate system. The radial coordinate, R , is transformed into a corrected radial coordinate, r :

$$r = R + AR^3 + BR^5$$

The A and B coefficients are determined by taking an image of a tape measure or similar device that spans the entire field of view and passes through the center pixel of the CCD camera. The pixel coordinates of marks on the tape measure in the image are converted to radial distances from the center pixel on the chip surface. A curve fit is made of the observed radii versus what the radii should be for an ideal undistorted image. The curve fit supplies the A and B coefficients.

The inverse transformation, if required, is given by:

$$R = r + Cr^3 + Dr^5$$

The C and D coefficients are determined with the same data used to determine A and B . The dependent and independent variables in the curve fitting are reversed in the procedure to determine the values of C and D .

B.2 Data Collection

It is necessary to identify the pixel that the optical axis of the lens intersects to obtain good data for curve fitting.

B.2.1 Center Pixel Determination. The *center pixel* is defined as the pixel nearest to the point on the surface of the detector through which the optical axis of the lens passes, as it is about this point that the distortions are radially symmetric. This is not necessarily the pixel nearest the center of the active elements in the detector array.

The identity of the center pixel can be determined by shining a laser into the camera lens. The laser should be fitted with a white screen several inches across with a small hole in the center of the screen to allow the beam to pass through the screen. When the laser is positioned along the optical axis of the camera lens, all the reflections from the optical elements inside the lens will be reflected back and centered on the laser aperture. While maintaining the laser's position, an image can be captured using a combination of filters and lens f -stop adjustments to prevent overexposure. The location of the laser light in the image reveals the location of the center pixel.

B.2.2 Measurement Procedure. A ruler, tape measure, or some other device with regular length markings is placed on a flat surface. In order to obtain optimal data, the camera shall be positioned so that a number of conditions are satisfied:

- The image of one of the markings on the tape measure is located at the center pixel.
- The optical axis of the lens shall be perpendicular to the measurement surface.
- The image of the tape measure should extend into the corner of the image farthest away from the center pixel.
- The camera is close enough to the tape measure to resolve the markings on it.

An exposure is made of the measuring device by the camera with an exposure time sufficient to capture the markings on the device. The distance between the camera lens and the measurement plane shall be noted.

B.3 Data Reduction

The purpose of taking an image of the tape measure is to create a dataset from which curves can be fit. The dataset will consist of pairs of numbers. One value of a pair is the computed undistorted location of an object in the image plane of the lens. The other value is the actual location of the pixel that corresponds to that object.

B.3.1 Actual Pixel Positions. It will be necessary to know the actual height, h , along a column and the width, w , across a row of the individual pixels. A list is made of the tape measure marks and their pixel coordinates. The radial distance, R , of a particular pixel away from the center of the center pixel can be calculated:

$$R = \left\{ \left[h(\text{Column} - \text{Column}_0) \right]^2 + \left[w(\text{Row} - \text{Row}_0) \right]^2 \right\}^{0.5},$$

where the coordinates of the center pixel are $(\text{Column}_0, \text{Row}_0)$, and the coordinates of the pixel of interest are $(\text{Column}, \text{Row})$.

B.3.2 Computed Pixel Positions. The undistorted positions of the tape measure marks in the image plane can be computed from:

$$r = \frac{df}{D},$$

where:

- d is the distance in the object plane from the center
- f is the focal length of the lens
- D is the distance between the object plane and the camera lens principal plane

The thin-lens approximation of geometrical optics assumes the lens to have no thickness and the principle optical planes coincide with the location of the lens. However, camera lenses are often complicated, multi-lens optical devices, and their behavior cannot be accurately modeled with the thin-lens approximation. The principle plane of a short-focus camera lens is

generally some distance in front of the lens. Therefore, it is difficult to determine the actual value of D without a trial-and-error process during curve fitting.

B.3.3 Curve Fit. Since the actual location of the principle plane of the camera lens is not generally known in advance, it is advisable to make the object-lens distance, D , a variable that is optimized to produce the smallest curve fitting errors. A curve fit can be made with an initial estimate of D . The value of D should be adjusted to minimize the average of the absolute values of the differences between the polynomial calculated distances and the distances computed from geometrical optics.

If the coefficients for the inverse transformation are also being computed, the average of the absolute values of the differences between the calculated distances and the actual pixel positions are also computed. The value of D should be adjusted to minimize the sum of the two sets of errors in this case.

Annex C – Noise Estimation

The estimation of uncertainty is an important aspect of any measurement. The concepts introduced in **Section 2.1** to compute the uncertainty of a CCD based measurement are described below. The uncertainty of a dark-current-corrected measurement will be the sum of the uncertainty of the uncorrected measurement and the uncertainty of the dark current. The CCD user will have knowledge of the number of bits; however, calculation of noise requires knowledge of the number of electrons collected in the pixel. If the uncorrected bit count is U_b , and the dark current bit count is D_b , then the number of electrons that generate the uncorrected bit count, U_e , is given by:

$$U_e = U_b \times \text{Gain},$$

where Gain is conversion factor between electrons and bits. The units of Gain are electrons/ADU. It is important to note that the term ADU (analog-to-digital unit) is often used in place of bits.

The number of electrons that generated the dark current, D_e , is given by:

$$D_e = D_b \times \text{Gain}$$

The noise in U_e , N_u , is given by:

$$N_u = (U_e + N_r^2)^{0.5},$$

where N_r is the read noise (number of electrons).

The noise in D_e , N_d is given by:

$$N_d = (D_e + N_r^2)^{0.5}$$

The noise in the corrected measurement, N_c , will be given by:

$$N_c = (N_u^2 + N_d^2)^{0.5}$$

If the number of pixels or exposures are averaged, the resulting noise in the averaged measurement, N' , is given by:

$$N' = \frac{N}{(XN)^{0.5}},$$

where:

X is the number of pixels or exposures

N is the noise in a single measurement

The percent uncertainty of a measurement with noise can be calculated from:

$$\%U = \frac{N}{M}, \text{expressed as a percentage,}$$

where:

$\%U$ = the percent uncertainty

M = the measurement, in number of electrons

N = the noise, in number of electrons

It should be noted that this derivation assumes that the CCD camera is operating normally and the light source is emitting a constant flux. If for any reason the camera is not operating normally, the actual noise production can vary from what is calculated. The determination of the actual noise has to be done statistically.

Annex D – Linearity Calculation Details

The linearity limits are determined by making a plot of bit count versus exposure time, with bit count on the y-axis and exposure time on the x-axis (see **Figure D-1**). A linear least-squares fit is then applied to the data. This will be of the form $y = mx + b$. If no latency is expected, then the fit to the data may be of the form $y = mx$. If latency is expected, the latent exposure time is given by $-b/m$. It is permissible to exclude data from the ends of the range to prevent it from affecting the fit.

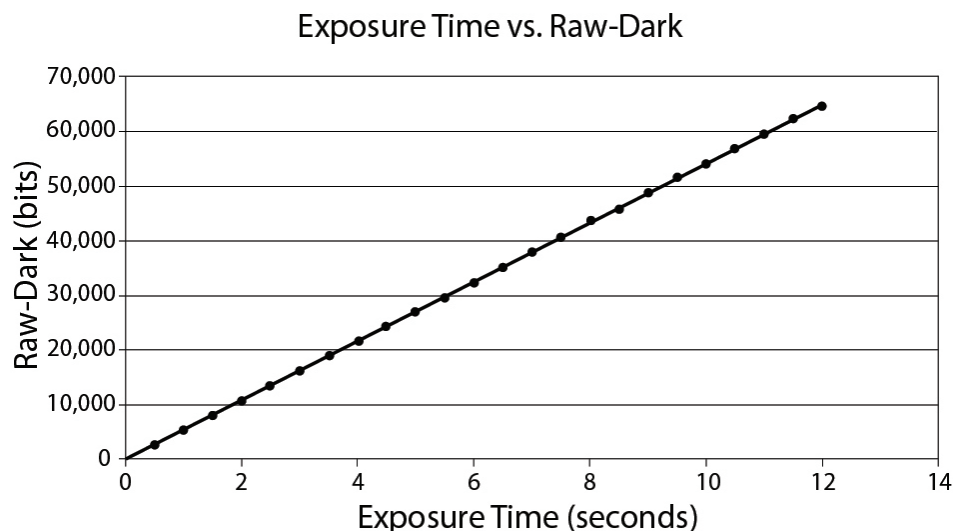


Figure D-1. Actual linearity data. (© Illuminating Engineering Society)

The difference, D , between the linear fit, F , and a measurement, M , of a given exposure time is calculated from:

$$D = M - F,$$

where:

$$F = m(\text{exposure time}) + b$$

The percent deviation from linearity, $\%D$, is calculated:

$$\%D = \frac{D}{F}, \text{ expressed as a percentage}$$

The percent deviation is calculated for all the exposure times. The percent deviation versus bit count is plotted on a graph with appropriately sized error bars for each point (see **Figure D-2**). (Refer to **Annex C** for error estimation.) The linear limits of the bit count range will be between the smallest and largest bit counts for which the percent deviation is within $\pm 1\%$, taking into account the error bars.

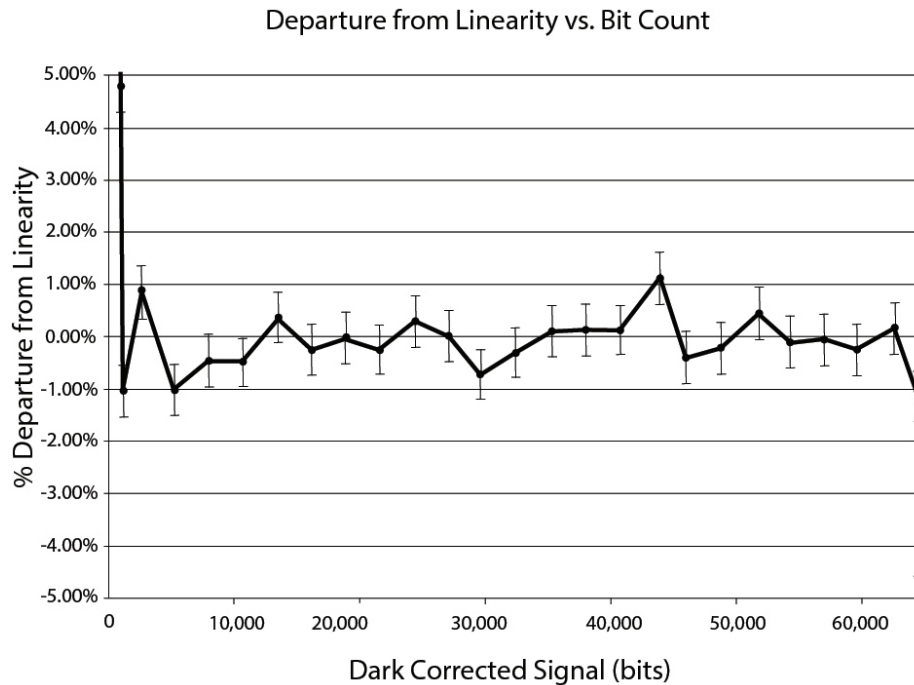


Figure D-2. Percentage departure from linearity vs. dark-corrected signal. (© Illuminating Engineering Society)

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